Coleman and Higher Coleman theories

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1 Introduction

1.1 Modular forms and congruences between them

Recall that we define the upper half plane $\mathfrak{h} := \{\tau \in \mathbb{C} | \mathrm{Im}(\tau) > 0\}$ with an action of $SL_2(\mathbb{Z}) =: \Gamma$. Given $k \in \mathbb{Z}$, a modular form of weight k and level Γ is an holomorphic function $f : \mathfrak{h} \to \mathbb{C}$ such that:

1. $\forall \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma$, we have $f(\gamma\tau) = (c\tau + d)^k f(\gamma)$.

2. f is holomorphic at ∞: For $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in \gamma$, we have $f(\tau+1) = f(\tau)$ so f has a Fourier expansion $f(\tau) = \sum_{i \in \mathbb{Z}} a_n q^n$, $q = e^{2\pi i \tau}$. Then f holomorphic at ∞ just means $a_n = 0$ for all $n < 0$. (Similarly, we have the notion of cusp form if f vanishes at ∞, i.e. $a_n = 0$ for $n \leqslant 0$).

We denote $M_k(\Gamma)$ the $\mathbb{C}$-vector space of modular forms of weight k and level Γ. Remark that they're finite dimensional vector spaces, and we have the action of Hecke operators T_n on them. The most important examples to keep in mind are the following ones:

Example 1. *1. $k \in \mathbb{Z}$, $G_k(\tau) = \frac{-B_k}{2k} + \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n$. B_k is the k-th Bernoulli number $\in \mathbb{Q}$, and $\sigma_{k-1}(n) = \sum_{d|n, d>0} d^{k-1}$.*

2. $\Delta = q \prod_{n=1}^{\infty} (1-q^n)^{24} = \sum_{\tau(n) q^n} \in S_k(\Gamma) \subseteq M_k(\Gamma)$.

Fix $p > 2$. Then, we have the following congruences, which imply congruences between q-expansions of modular forms.

Proposition 1. *(Kummer's congruences) $\frac{B_k}{2_k}$ is a p-integer (i.e. $\in \mathbb{Z}_{(p)}$) if $k \not\equiv 0 \mod p-1$, and, if $k \equiv k' \mod p-1$, then we have*

$$\frac{B_k}{2k} \equiv \frac{B_{k'}}{2k'} \mod p$$

In particular, this implies that $G_k \equiv G_{k'}$, seen as series in $\mathbb{Z}_{(p)}[[q]]$.

There are also more congruences of this kind for $k \equiv k' \mod p^n(p-1)$.

Now, ovserve that, in particular, if $(p-1)|k$, then

$$E_k = 1 - \frac{2k}{B_k} \sum_{n=1}^{\infty} \sigma_{k-1}(n) q^n \implies E_k \equiv 1 \mod p$$

Given these congruences, it's natural to ask the following question.

Question: Do congruences between modular forms imply that they live in p-adic families indexed by the weight?

More generally, for $N \geqslant 1$, let $\Gamma_1(N) = \Gamma \subseteq SL_2(\mathbb{Z})$ and $k \in \mathbb{Z}$. We have the vector spaces $M_k(\Gamma_1(N))$ and cusps $\mathbb{P}^1(\mathbb{Q})$. We also have Hecke operators T_p for $p \nmid N$, U_p for $p \mid N$ and $\langle a \rangle$ for $a \in (\mathbb{Z}/N\mathbb{Z})^{\times}$. We want to know if we can see some p-adic variation of the family $\{M_k(\Gamma_1(N))\}_{k \in \mathbb{Z}}$.

This was first done by Hida (80', for ordinary modular forms) and later by Coleman (90', for finitye slope modular forms). To vary modular forms p-adically, we need two things:

1. Some algebraicity property of modular forms.
2. p-adic variation of this algebraic incarnation.

in the next section we will see an overview of how (1) and (2) can be done.

1.2 Families of modular forms

For (1), we need to see **modular forms geometrically**. That is, we want to see modular forms (+ Hecke operators) as classes in cohomology groups (coherent cohomology of schemes, or Betti cohomology of some manifold). It should be pointed out that not all forms can be seen in this way, see for instance Maß forms.

As an example, fix $\Gamma = \Gamma_1(N)$ and $N \geqslant 4$. Let $Y_1(N)$ be the moduli space of elliptic curves E with level $\Gamma_1(N)$-structure $(\mu_n \hookrightarrow E[N])$. Then, $Y_1(N)$ is a curve over $\mathrm{Spec}(\mathbb{Z}[1/N])$, smooth, affine. We have an universal elliptic curve $E \xrightarrow{\pi} Y_1(N)$ and a section $e : Y_1(N) \to E$. We also have a universal line bundle $\underline{\omega} := \pi_*(\Omega^1_{E/Y_1(N)}) = e^*(\Omega^1_{E/Y_1(N)})$.

If we base-change it to $\mathbb{C}$, we get $Y_1(N)_{\mathbb{C}} = Y_1(N) \times_{\mathrm{Spec}(\mathbb{Z}[1/N])} \mathrm{Spec}\,\mathbb{C} \cong \Gamma_1(N) \backslash \mathfrak{h}$. There is also a natural compactification $X_1(N) \supseteq Y_1(N)$ such that $X_{\mathbb{C}} = \Gamma_1(N) \backslash (\mathfrak{h} \cup \mathbb{P}^1(\mathbb{Q}))$.

Moreover, we have a commutative diagram

$$\begin{array}{ccc} E & \hookrightarrow & \widetilde{E} \\ {\scriptstyle e}\uparrow\downarrow{\scriptstyle \pi} & & {\scriptstyle \tilde{e}}\uparrow\downarrow \\ Y_1(N) & \hookrightarrow & X_1(N) \end{array}$$

$\widetilde{E}$ is a semi-abelian scheme, and $\underline{\omega}$ extends canonically to X as $\tilde{e}^*(\Omega^1_{\widetilde{E}/X})$.

Proposition 2. *Over $\mathbb{C}$, for every $k \in \mathbb{Z}$, there is an isomorphism (as Hecke modules)*

$$M_k(\Gamma_1(N)) \cong H^0(X_{\mathbb{C}}, \underline{\omega}_{\mathbb{C}}^{\otimes k})$$

In particular, for $X \to \mathrm{Spec}(\mathbb{Z}[1/N])$, we may define classical modular forms of weight k over $\mathbb{Q}_p$, by considering $M_k(\Gamma_1(N), \mathbb{Q}_p) = H^0(X_{\mathbb{Q}_p}, \underline{\omega}_{\mathbb{Q}_p}^{\otimes k})$. Note that these are finite dimensional $\mathbb{Q}_p$-vector spaces.

Now for (2), we give a geometric approach, although it can also be done using q-expansions, as we will see in the future. First we need a parameter space over which we can vary the sheaves of modular forms. We can think of an integer k as a map $k : \mathbb{G}_m \to \mathbb{G}_m$, which on points is given by $k : \mathbb{G}_m(R) \to \mathbb{G}_m(R)$, $x \mapsto x^k$. Now we replace $\mathbb{G}_m$ by $\mathbb{Z}_p^* \subseteq \mathbb{G}_m$.

If R is a topological $\mathbb{Q}_p$-algebra, we can consider $W(R) = \mathrm{Hom}_{cont}(\mathbb{Z}_p^*, R^*)$. W is a rigid space, rigid generic fiber of $\mathrm{Spf}(\Lambda)$, for $\Lambda = \mathbb{Z}_p[[\mathbb{Z}_p^*]]$. W is a disjoint union of $(p-1)$ open unit balls $\mathbb{A}^1_{\mathbb{Q}_p}$, and it contains the classical weights $k \in \mathbb{Z}$: Observe that $W(\mathbb{Q}_p) = \mathrm{Hom}_{cont}(\mathbb{Z}_p^*, \mathbb{Q}_p^*) \cong \mathrm{Hom}_{cont}(\mathbb{Z}_p^*, \mathbb{Z}_p^*)$, and each $(x \mapsto x^k) \in W(\mathbb{Q}_p)$.

Now that we have the parameter space, we want to interpolate p-adically $\{\underline{\omega}^k\}_{k\in\mathbb{Z}}$, for $\underline{\omega}$ line bundle on X. That is, for $\kappa \in W(\mathbb{Q}_p)$, we would like to define $\underline{\omega}^\kappa$.

Now, let $X_1(N)$ over $\mathbb{Z}[1/N]$. For $p \nmid N$, we can define the reduction modulo p, $X_1(N)_{\overline{\mathbb{F}_p}}$, and the rigid analyfication $X_1(N)^{an}_{\mathbb{Q}_p}$, and we have a projection $X_1(N)^{an}_{\mathbb{Q}_p} \to X_1(N)_{\overline{\mathbb{F}_p}}$. Then, we have in $X_1(N)_{\overline{\mathbb{F}_p}}$ three types of points: Those corresponding to cusps, those corresponding to ordinary elliptic curves, and those corresponding to supersingular elliptic curves. For each supersingular point in $X_1(N)_{\overline{\mathbb{F}_p}}$, its preimage in $X_1(N)^{an}_{\mathbb{Q}_p}$ are some open disks, and there are finitely many of them, as it is shown on Figure 1.

Figure 1: Supersingular disks

Definition 1. *If we remove the supersingular disks in $X_1(N)^{an}_{\mathbb{Q}_p}$, we obtain the ordinary locus X^{ord}.*

$$X^{ord} = X \backslash \bigcup \textit{supersingular disks}$$

Definition 2. *Instead of removing all supersingular disks, we remove some open subdisk, and we get some strict neighborhoods of X^{ord} in X. Then, we define*

$$X^{ord,\dagger} = \varinjlim \textit{ strict neighborhoods of } X^{ord} \textit{ in } X$$

Goal: $\forall \kappa \in W(\mathbb{Q}_p)$, we wish to define a line bundle $\underline{\omega}^\kappa$ on $X^{ord,\dagger}$, such that, if $\kappa = k \in \mathbb{Z}$, then $\underline{\omega}^\kappa = \underline{\omega}^k|_{X^{ord,\dagger}}$. As a first approximation, we define $\underline{\omega}^\kappa$ on X^{ord}.

Let E be an ordinary elliptic curve over $K/\mathbb{Q}_p$. We have the Tate module $T_pE = \varprojlim E[p^n]$ with a $G_K = \mathrm{Gal}(\overline{K}/K)$-action. We have ω_E the universal differential. Recall that we have a Hodge-Tate map

$$T_pE \otimes_{\mathbb{Z}_p} \mathbb{C}_p \to \omega_E \otimes \mathbb{C}_p$$

Moreover, as E is ordinary, we have an exact sequence

$$0 \to T_pE^m \to T_pE \to T_pE^{\acute{e}t} \to 0$$

And the Hodge-Tate map gives an isomorphism $T_pE^{\acute{e}t} \otimes_{\mathbb{Z}_p} \mathbb{C}_p \cong \omega_E \otimes_K \mathbb{C}_p$, compatible with the G_K-action.

Now let $\kappa \in W(\mathbb{Q}_p)$. We define $((T_pE)^{\acute{e}t})^\kappa := \mathbb{Z}_p$ with an action given by $G_K \xrightarrow{\rho} \mathbb{Z}_p^* \xrightarrow{\kappa} \mathbb{Z}_p^*$, and we let

$$\omega_E^\kappa = \left((T_pE^{\acute{e}t})^\kappa \otimes_{\mathbb{Z}_p} \mathbb{C}_p\right)^{G_K}$$

This works for families of ordinary elliptic curves $E^{ord} \to X^{ord}$, and allows to define ω^κ on X^{ord}. It turns out, however, that this is not good enough for some reasons, and we need overconvergence.

1.3 Overconvergence

The procedure that we followed above doesn't work in the overconvergent case $X^{ord,\dagger}$, because when we cross the singular locus, T_pE is irreducible, so we don't have the étale quotient at our disposal.

Now, if we want overconvergence, we replace $\mathbb{Z}_p^*$ by some thickening of it. $\kappa \in W(\mathbb{Q}_p)$, $\kappa : \mathbb{Z}_p^* \to \mathbb{Z}_p^*$ continuous, it is locally analytic, so there exists a strict neighborhood $X^{ord} \subseteq U \subseteq X$ such that $U = \mathrm{Spa}(A, A^+)$ is an affinoid,

and there is some $n \gg 0$ such that the thickening $\mathbb{Z}_p^*(1+p^nA^+) \ni x$, $\kappa(x)$ makes sense, and so we can define $\underline{\omega}^\kappa$ on the strict neighborhood U.

Then, we have

$$\begin{array}{ccc} \text{classical weights} = \text{characters of } \mathbb{G}_m & \longrightarrow & \text{automorphic sheaves } \underline{\omega}^k \\ \downarrow & & \\ \text{p-adic weights} = \text{characters of } \mathbb{Z}_p^* & \longrightarrow & \underline{\omega}^\kappa \text{ line bundles on } X^{ord,\dagger} \end{array}$$

Definition 3. *Given $\kappa \in W(\mathbb{Q}_p)$, $M_k^\dagger(\Gamma_1(N)) := H^0(X^{ord,\dagger}, \underline{\omega}^\kappa) \circlearrowleft$ Hecke operators, U_p compact. It is an inductive limit of p-adic Banach spaces.*

We also define the slope as the valuation of the generalized eigenvalues.

Given a classical weight $k \in \mathbb{Z}$, we have

$$H^0(X, \underline{\omega}^k) \to H^0(X^{ord,\dagger}, \underline{\omega}^\kappa) \circlearrowright U_p$$

On $H^0(X, \underline{\omega}^k)$ we don't have the operator U_p, so we replace X by $X' = X(\Gamma_1(N) \cap \Gamma_0(p))$. We have the following diagram

$$\begin{array}{ccc} X' = X(\Gamma_1(N) \cap \Gamma_0(p)) & & \\ \downarrow \pi & \nearrow & \\ X & \hookleftarrow & X^{ord,\dagger} \end{array}$$

The section $X^{ord,\dagger} \to X'$ allows to construct a map

$$\varphi : H^0(X', \underline{\omega}^k) \to H^0(X^{ord,\dagger}, \underline{\omega}^\kappa)$$

With commuting actions of the Hecke operators on both sides. We have the following result:

Theorem 1 (Coleman)**.** *φ is an isomorphism of slopes h on both sides, for $h < k-1$.*

This tells us that an overconvergent modular for of small slope is classical.

1.4 Higher Coleman theory

For now, we have discussed $H^0(X, \underline{\omega})$. What about the higher cohomology groups?

For $k \geqslant 2$, $H^1(X, \omega^k) = 0$, but for $k = 1$, we have, via cup product and the Kodaira-Spencer isomorphism,

$$H^0(X, \underline{\omega}) \times H^1(X, \underline{\omega}) \xrightarrow{\cup} H^1(X, \underline{\omega}^2) \cong H^1(X, \Omega^1_X(\log)) = K.$$

$k = 1$ is an irregular weight, we have an Artin Galois representation of Hodge-Tate weights (0,0). This is what we call an irregular motive.

2 Ordinary locus and its neighborhoods

References:

- Katz, p-adic properties of modular schemes and modular forms, [6] sections 2 and 3.
- Buzzard, Analytic continuation of overconvergent eigenforms, [1].

Remark 1. *A good observation to motivate what we are doing is the folllowing one in [3]. We would like to have p-adically a decomposition of $M_k(\Gamma, R)$ into an infinite sum of eigenspaces, as we have over $\mathbb{C}$. However, this is not possible because U_p has a continuous spectrum. For this we consider sections of X which converge beyond the ordinary locus (we will see that in the next section).*

Now, an ordinary elliptic curve comes with a canonical subgroup (the subgroup scheme given by the kernel of reduction $\mod p$). This allows us to go from level 1 to level $\Gamma_0(p)$. We want to generalize this construction to elliptic curves which are no longer ordinary.

Let p prime, $(N,p) = 1$ and let $Y = Y_1(N)$ be the modular scheme over $\mathbb{Z}[1/N]$. Let $X = X_1(N)$ be its compactification, $W := W(\overline{\mathbb{F}_p}) = \widehat{\mathbb{Z}_p^{ur}}$ Witt vectors, and $K_0 = W[1/p]$. K/K_0 finite extension. We consider the reduction modulo p of the compactified modular curve X, red : $X(K) \to X(\mathbb{F}_p)$. For $x \in X(\overline{\mathbb{F}_p})$, $\mathscr{O}_{X,x} \cong W[[T_x]]$. We fix such an isomorphism for every supersingular point x. This gives an explicit isomorphism $\mathrm{red}^{-1}(\{x\}) =: B_x \cong \mathbb{D}(0,1)$ open unit disk. Now we're working over a rigid space.

Definition 4. *Let $r = p^q \in (0,1]$, for some $q \in \mathbb{Q}$. Then we define $B_x^{\leqslant r} := \{p \in B_x \text{ such that } |T_x(p)| < r\}$. Moreover, we let $X_{\geqslant r} = X^{an} \backslash \bigcup_{x \text{ supersingular}} B_x^{\leqslant r}$.*

In particular, $X_{\geqslant 1} =: X^{ord}$ is the ordinary locus, and $\{X_{\geqslant r_0}\}$ is a system of strict neighborhoods of X^{ord}.

Remark 2. *Any automorphism $W[[T]] \to W[[T]]$ is given by $T \mapsto pa + Tu(T)$, with $a \in W$ and $u(T) \in W[[T]]$ a unit.*

Definition 5. *Given a point $\alpha = (E,i) \in X(\overline{K})$, we define*

$$v(E,i) = \begin{cases} 0 & \text{if } E \text{ has multiplicative or good ordinary reduction} \\ v(T_x(\alpha)) & \text{if } \alpha \text{ reduces to } x \mod p \end{cases}$$

This measures in some sense "how supersingular the curve is", as next definition illustrates.

Definition 6. *Let L/K be a finite extension, and E/L elliptic curve. We say that E is too supersingular if for some $i : \mu_N \to E$, $v(E,i) \geqslant \frac{p}{p+1}$. Otherwise E is said to be not too supersingular.*

This is well defined because it turns out to be independent of i. We denote from now on simply $v(E)$.

Remark 3. *Observe that T_x is a parameter on the disk of points above $x \in X_{\overline{\mathbb{F}_p}}$. Then, $|(E,i)| = p^{-v(E,i)}$, so not too supersingular curves are those close to the edge of the supersingular disk B_x.*

To an elliptic curve that is not too supersingular, we can associate a canonical subgroup.

Theorem 2 (Katz, Lubin)**.** *For E/L not too supersingular, there exists $H := H(E) \subset E[p]$, a canonical subgroup of order p. Moreover, we have the following properties:*

(a) If $v(E) = 0$, $H = \ker(E[p](\overline{L}) \to \mathscr{E}_{\mathbb{F}_p}(\overline{\mathbb{F}_p}))$, where $\mathscr{E}$ is the Néron model of E over $\mathscr{O}_L$.

(b) If $v(E) \leqslant \frac{1}{p+1}$, then $v(E/H) = pv(E)$, and in particular E/H is not too supersingular.

(c) If $v(E) = \frac{1}{p+1}$ then E/H is too supersingular.

(d) If $v(E) \in \left(\frac{1}{p+1}, \frac{p}{p+1}\right)$, then $v(E/H) = 1 - v(E)$, E/H is again not too supersingular and its canonical subgroup is $E[p]/H$.

(e) If $v(E) < \frac{p}{p+1}$ and $C \subset E[p]$, $C \neq H$, then $v(E/C) = v(E)/p$, and $H(E/C) = E[p]/C$.

(f) If $v(E) \geqslant \frac{p}{p+1}$, $C \subset E[p]$ subgroup of order p, then $v(E/C) = \frac{1}{p+1}$.

The objective of this lecture is to study the construction of the canonical subgroup H.

Remark 4. *Property (e) above is very important: As the U_p operator is defined as the sum over E/C, for C not the canonical subgroup, (e) can be phrased as "the U_p operator increases overconvergence".*

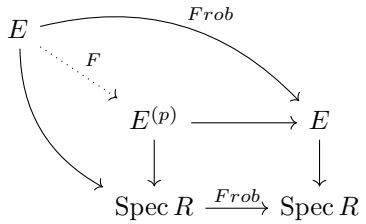
2.1 The Hasse invariant

Let $E \xrightarrow{\pi} X = X_1(N)$ universal generalized elliptic curve, e a section, $\omega = e^*\Omega^1_{E/X}$. $M_k(\Gamma_1(N), \overline{K}) = H^0(X_{\overline{K}}, \omega_R^{\otimes k})$.

$$\begin{array}{ccc} E|_Y & \hookrightarrow & E^{\text{univ}} \\ \downarrow & & \downarrow \pi \\ Y & \hookrightarrow & X = X_1(N) \end{array}$$

For an elliptic curve E/R, $\Omega^1_{E/R} = \mathscr{O}_E\omega$ is free, we have $\begin{array}{ccc} E & \hookrightarrow & E^{\text{univ}} \\ \downarrow & & \downarrow \\ \operatorname{Spec} R & \xrightarrow{t} & Y \end{array}$. Let $t^*(f) \in H^0(E, \Omega^{\otimes k}_{E/R}) \cong R\omega^{\otimes k}$.

For $\lambda \in R^\times$, $f(E, \lambda\omega)(\lambda\omega)^{\otimes k} = f(E,\omega)\omega^{\otimes k}$. Now let R be some $\mathbb{F}_p$-algebra, E/R elliptic curve, we have



Pick $\eta \in H^1(E, \mathscr{O}_E)$ basis, dual to ω basis of $H^0(E, \Omega^1_{E/R})$ (observe that $H^1(E, \mathscr{O}_E)$ is the dual of $H^0(E, \Omega^1_{E/R})$ by Serre duality). The absolute Frobenius $\text{Frob}_E : E \to E$ induces an $\mathbb{F}_p$-linear morphism

$$\text{Frob}_E^* : H^1(E, \mathscr{O}_E) \to H^1(E, \mathscr{O}_E)$$

Observe that as η is a basis, we must have

$$\text{Frob}_E^*(\eta) = A(E/R, \omega)\eta$$

Definition 7. *We define the Hasse invariant $A(E/R,\omega)$ given by $Frob_E^*(\eta) = A(E/R,\omega)\eta$. It turns out it is a modular form of weight $p-1$. Indeed, if we take, for $\lambda \in R^\times$ a basis $\lambda\omega$ instead of ω, we get the dual basis $\lambda^{-1}\eta$, and we have*

$$Frob_E^*(\lambda^{-1}\eta) = \lambda^{-p} Frob_E^*(\eta) = \lambda^{-p}A(E/R,\omega)\eta = \lambda^{1-p}A(E/R,\omega)(\lambda^{-1}\eta)$$

Remark 5. *We can define q-expandions given by evaluation at the Tate curve:*

$$f(Tate(q)/R((q)), \omega_{can}, i) \in R((q))$$

And, in fact, by the homolorphicity condition at the cusps it lands on $R[[q]]$. The Hasse invariant is such that $A(Tate(q), \omega_{can}) = 1$.

Remark 6. *Let $p \geqslant 5$, $E_{p-1}(q) = 1 - \frac{2(p-1)}{B_{p-1}} \sum \sigma_{p-2}(n)q^n$. We have E_{p-1} is a modular form of weight $p-1$, and by Kummer congruences, we have that, as power series in $R[[q]]$*

$$E_{p-1}(q) \equiv A(q) \mod p$$

Remark 7. *A has simple zeroes at every supersingular point (c.f [7], Theorem 12.4.3). Moreover, for every supersingular point x, we pick U such that $\omega^{\otimes p-1}|_U \cong \mathscr{O}_U$, and let f_x be that trivializing isomorphism, under which 1 corresponds to some ω_0. Then we can pick as uniformizing parameter $T_x = f_x(E_{p-1})$, and if $\alpha = (E,i)/L$ is such that $red(E,i) = x$, then we have $v(E,i) = v(T_x(\alpha)) = v(E_{p-1}(E,\omega_0))$. This makes it clear that $v(E,i)$ does not depend on the level structure.*

Now, the construction of the canonical subgroup is based on the study of the formal group of the elliptic curve. This follows closely [6], section 3.6.

2.2 Generalities on formal groups

Definition 8. *A (one parameter, commutative) formal group law over a ring R is a power series $G(X,Y) \in R[[X,Y]]$ satisfying the following properties:*

1. $G(G(X,Y),Z) = G(X,G(Y,Z))$
2. $G(X,Y) = G(Y,X)$
3. $G(X,Y) = X + Y +$ *higher order terms*

A morphism of formal groups over R, $f : G \to H$ is $f \in R[[T]]$ satisfying that $f(G(X,Y)) = H(f(X), f(Y))$.

Example 2.
- $\mathbb{G}_a(X,Y) = X + Y$ *additive formal group.*
- $\mathbb{G}_m = X + Y + XY$ *multiplicative formal group.*
- *Given an elliptic curve E/R, take $\widehat{E}/R$ the formal completion of E along the identity section, $\widehat{E} \cong Spf(R[[T]])$*

Lemma 1. *Let R be a $\mathbb{Z}_p$-algebra, G/R a formal group. Then*

1. $\mathrm{End}_R(G)$*, which is given by power series $f \in R[[T]]$ such that $f(G(X,Y)) = G(f(X), f(Y))$ satisfies $Z(\mathrm{End}_R(G)) \supseteq \mathbb{Z}_p$, where Z denotes the center.*
2. *Given a parameter X_0, there exists a non-unique parameter $X \equiv X_0 \mod X_0^2$ such that $\zeta \in \mu_{p-1} \subset \mathbb{Z}_p$ satisfies $[\zeta](X) = \zeta X$. We call such a parameter p-typical.*

Lemma 2. *Let R be a $\mathbb{F}_p$-algebra, G/R be a formal group for any parameter X. Then, $[p](X) = V(X^p) = \sum_n v_n X^{np}$, where multiplication by p factors as $[p] = V \circ F$, $F : G \to G^{(p)}$ given by $X \mapsto X^p$.*

Lemma 3. *Let R be a Z_p-algebra, G/R be a formal group, and X a p-typical parameter. Then, $[p](X) = Xf(X^{p-1})$ for some $f \in R[[X]]$.*

Lemma 4. *Let R be a Z_p-algebra, X any parameter, G/R a formal group. Let $\omega(X) = (1 + \sum_n a_n X^n)\, dX$ be an invariant differential, i.e. satisfying $\omega(G(X+Y)) = \omega(X) + \omega(Y)$. Then, we have that*

$$[p](X) \equiv a_{p-1} X^p \mod (p, X^{p+1})$$

2.3 Formal group law of an elliptic curve

Lemma 5. *Let R be a $\mathbb{F}_p$-algebra, E/R an elliptic curve, ω an invariant differential and X a choice of parameter or $\hat{E}$ such that $\omega(X) = (1 + \sum_n a_n X^n)\, dX$. Then, if $A(E,\omega)$ denotes the Hasse invariant, we have*

$$a_{p^n-1} = A(E,\omega)^{\frac{p^n-1}{p-1}}$$

Proof. $V^* : \Omega^1_{E/R} \to \Omega^1_{E^{(p)}/R} = (\Omega^1_{E/R})^{(p)}$, $V^*\omega = A(E,\omega)\omega^{\otimes p}$ is given by the Cartier operator:

$$C(X^n dx) = \begin{cases} 0 & \text{if } p \nmid n+1 \\ (X^{\frac{n+1}{p}-1} dX)^{(p)} & \text{if } p \mid n+1 \end{cases}$$

Therefore, $C(\omega) = \sum_{n \geqslant 0} a_{p(n+1)-1} (X^n dX)^{(p)} = A(E,\omega)\left(1 + \sum_{n \geqslant 1} a_n X^n dX\right)^{(p)} = A(E,\omega)\left(1 + a_n^p (X^n dX)^{(p)}\right)$. And so we have $a_{p(m+1)-1} = A(E,\omega) a_m^p$, and letting $m + 1 = p^{n-1}$ we get $a_{p^n-1} = A(E,\omega) a_{p^{n-1}-1}^p$. Now using that $a_0 = 1$ we get the desired result (if you want, by induction on n). □

The following proposition summarizes the results we need from the past lemmas:

Proposition 3. *Let R be a $\mathbb{Z}_p$-algebra, E/R an elliptic curve, ω an invariant differential and X a p-typical parameter for the formal group such that $\omega = \left(1 + \sum_{n \geqslant 1} a_n X^n\right) dX$. Then,*

$$[p](X) = pX + aX^p + \sum_{m \geqslant 2} c_m X^{m(p-1)+1}$$

with $c_m \equiv 0 \mod p$ unless $m(p-1) + 1 \equiv 0 \mod p$ (that is, unless $m \equiv 1 \mod p$). Moreover, $a \equiv A(E,\omega) \mod p$.

Proof. By 3 we have that $[p](X) = Xf(X^{p-1})$, so this justifies that we only have the terms $(p-1)m+1$. Moreover, by 4 we have that modulo p, $[p](X)$ has to be a series on X^p, so this implies the congruences for c_m. Finally, 5 implies the congruence of a with the Hasse invariant. □

Now, we want to make a choice of the canonical subgroup inside $\ker[p]$. This is equivalent to picking a $\mathbb{F}_p$-space of dimension 1, which is equivalent to pick a $\mathbb{F}_p^\times$-invariant subspace. The action of $\mathbb{F}_p^\times$ is given by multiplication by ζ, $[\zeta](X) = \zeta X + \text{h.o.t.}$. Then, we can let $X^{p-1} = T$ and choosing a canonical subgroup corresponds to choosing canonically a root T of $g(T) = aT + \sum_{m\geqslant 2} c_r T^m$, where

$$[p](X) = Xg(T)$$

A canonical choice of such a root can be made by picking that of minimal valuation. For this we need Newton's polygons.

Let $f(X) = \sum_n a_r X^r \in R[[X]]$, with $R = \mathscr{O}_L$, L complete valued field. Suppose that $n = \min\{r \text{ such that } a_r \in R^\times\} \leqslant \infty$. Let $N(f)$ be the upper convex hull of the set $\{(i, v(a_i)\}$. For instance, if $f(x) = p^3 + p^4X + p^2X^2 + p^3X^3$ we have the following Newton polygon:

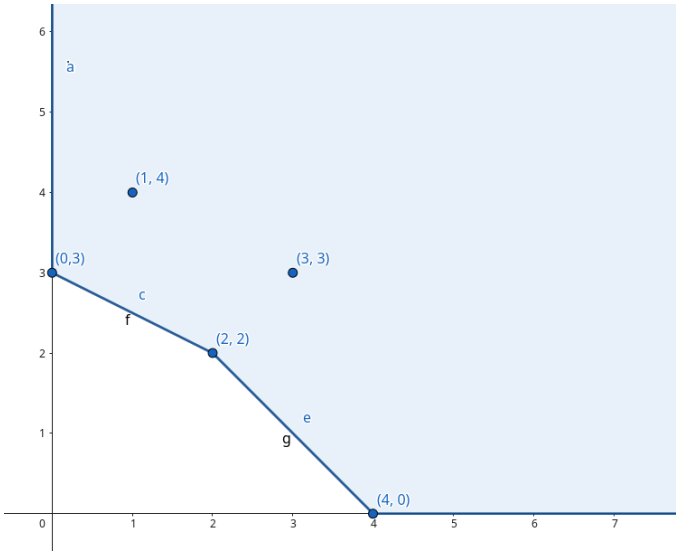


Figure 2: Example Newton polygon for $f(X) = p^3 + p^4X + p^2X^2 + p^3X^3$

Proposition 4 (Weierstrass Preparation)**.** *For R a complete and separated ring for the p-adic topology, $f \in R[[T]]$, there exists $g \in R[[X]]^\times$ and $h \in R[X]$ monic of degree n such that $f(X) = h(X)g(X)$. Moreover, if $h = \prod(X - \alpha_i)$ and $\mu_i = v(\alpha_i)$, with $\mu_1 \geqslant \mu_2 \geqslant \dots$, then the slopes of segments in $N(f)$ are $-\mu_1, -\mu_2, \dots$.*

Now, recall that we have E/R elliptic curve, $R = \mathscr{O}_L$, $v(E) = v(E_{p-1}(E,\omega)) < \frac{p}{p+1}$, so the curve is not too supersingular, and we have $[p](X)$ given as in 3. Now we have two cases:

- **Ordinary case:** $v(E) = 0$, $[p](X) = Xf(T)$, $f(T)$ factors as $(T - \alpha_{can})u(T)$, for $u(T)$ a unit in $R[[T]]$. Then,

  $$[p](X) = (X^p - X\alpha_{can})u(X^{p-1})$$

  and we can define $H(E) := Z(X^p - \alpha_{can}X)$, which is the kernel of the reduction $\mod p$.

- **Supersingular case:** In characteristic p, the height of G formal group is the largest h such that $[p](X) = f(X^{p^h})$. For an elliptic curve we have that

  $$ht(\hat{E}) = \begin{cases} 1 & \text{ordinary} \\ 2 & \text{supersingular} \end{cases}$$

  We have

  $$[p](X) = pX + aX^p + \sum_{m=2}^{p} c_m X^{m(p-1)+1} + c_{p+1}X^{p^2} + \sum_{m\geqslant p+2} c_m X^{m(p-1)+1} \tag{1}$$

with $v(E) = v(a)$, and then if $v(E) \leqslant \frac{p}{p+1}$, we can pick the $p-1$ non-zero solutions of $[p](X) = 0$ corresponding to valuation $\frac{1-v(a)}{p-1}$. The other solutions all have valuation $\frac{v(a)}{p^2-p}$, which is less than $\frac{1-v(a)}{p-1}$. If $v(E) \geqslant \frac{p}{p+1}$, then all non-zero solutions of $[p](X) = 0$ have valuation $\frac{1}{p^2-1}$, so we can't construct a canonical subgroup. This can be seen looking at the Newton polygon for 1:

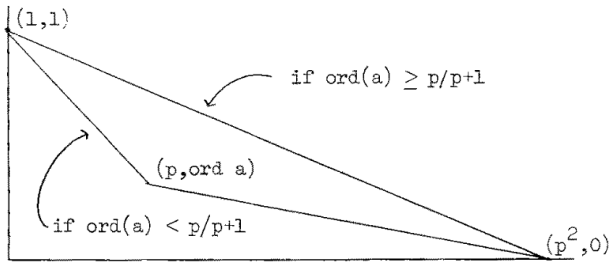


Figure 3: Newton polygon for $[p](X)$, extracted from [6], Theorem 3.10.7.

2.4 Geometry of $X_1(N, p)$

$X_1(N, p)$ is the completed modular curve of level $\Gamma_1(N) \cap \Gamma_0(p)$. The open modular curve $Y_1(N, p)$ parametrizes triples (E, i, C) with E/R elliptic curve, $i : \mu_N \to E[N]$ and $C \subseteq E[p]$. Its reduction modulo p has two components that intersect at supersingular points:

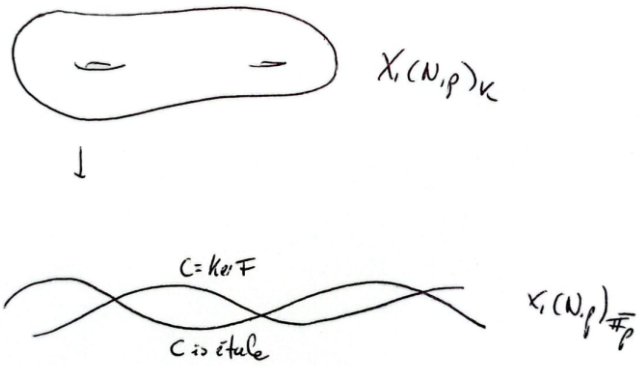


Figure 4: $X_1(N, p)$ and its reduction mod p

We have the natural maps $\pi_1, \pi_2 : X_1(N, p) \to X_1(N)$. The theory of the canonical subgroup provides π_1 with a section on $X_1(N)^{ord}$ that extends beyond the ordinary locus.

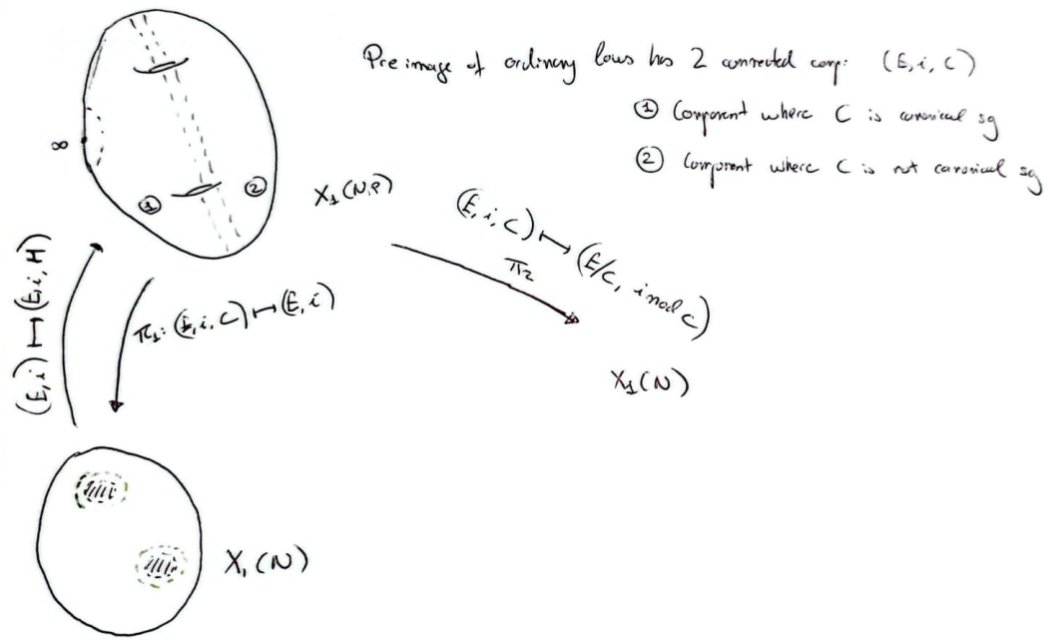


Figure 5: Maps between $X_1(N,p)$ and $X_1(N)$

3 Overconvergent modular forms and the spectral theory of the U_p operator

3.1 p-adic functional analysis

Definition 9. *Let K be a nonarchimedian field, p prime. A K-Banach space E is a normed vector space over K, such that $|x+y| \leqslant \max\{|x|,|y|\}$ for every $x, y \in E$.*

We require the following technical condition: N) If $|.| : K^\times \to G \subseteq \mathbb{R}_{\geqslant 0}$ we require that for every $x \in E$, $|x| \in \overline{G}$. It is always possible to find an equivalent norm such that this condition holds.

Example 3. *Let E be a set, we consider $c(I)$ the space of sequences $(x_i)_{i\in I}$ of elements of K such that $|x_i| \to 0$. It is a Banach K-space with the norm $|(x_i)| = \sup_{i\in I} |x_i|$.*

Definition 10. *Let E be a K-Banach space, $(e_i) \subseteq E$ is called an ON-basis if*

1. *$\forall x \in E$, $x = \sum c_i e_i$ uniquely, with $c_i \to 0$.*
2. *$|x| = \sup |c_i|$ for every $x \in E$.*

Remark 8. *E has an ON-basis (and is called ONable or orthonormalizable) if and only if $E \cong c(I)$ as a K-Banach space, that is, E is isometric to $c(I)$ for some I. It is interesting to extend the notion to include isomorphism instead of isometry. This is the following definition.*

Definition 11. *E is said to be potentially ON if $E \cong c(I)$ for some set I, as topological K-vector spaces.*

Lemma 6. *Let K discrete, E a K-Banach space. Then E is potentially orthonormalizable. Moreover, for K discrete and E a K-Banach space satisfying N), then (e_i) is an ON-basis of E if and only if $|e_i| = 1$, and $(\overline{e_i}) \subseteq E_0/\mathfrak{m}_K E_0$ gives an (algebraic) basis.*

Lemma 7 (Existence of projectors)**.** *Let E be a K-Banach space, K discrete. If F is a closed subspace of E, then there is a continuous projector $\Pi : E \to F$, and $E = F \times \ker \Pi$.*

Proof. E/F is potentially ON, let $(\overline{e_i})$ be a ON-basis of E/F, and lift it to (e_i) with $|e_i| \leqslant 1$. Then we can define $s : E/F \to E$ given by $\overline{e_i} \mapsto e_i$, which gives a section and so we get a splitting. □

3.1.1 Linear operators

Now we consider linear operators berween K-Banach spaces. If E, F are K-Banach spaces, we can define $\mathscr{L}(E,F)$ to be the set of continuous maps from E to F. It has a structure of K-Banach space with the norm $|u| = \sup \frac{|ux|}{|x|}$.

Remark 9. *Let $E = c(I)$, then $\mathscr{L}(E,F) \cong b_I(F)$ bounded sequences of elements of F. This is clear, as the image is determined by the image of the basis (e_i).*

In particular, if $F = c(J)$, then every element $\phi \in \mathscr{L}(E,F)$ is determined by a sequence $(w_j)_{j\in J}$ of elements $w_j \in E^* = \mathscr{L}(E,K)$, that is, we can see operators as "infinity size matrices".

Definition 12. *We say that $\phi \in \mathscr{L}(E,F)$ is a finite rank operator if its image has finite dimension in F, and we denote the set of such operators $\mathscr{L}(E,F)^{\mathrm{fin}}$. We say that u is a completely continuous operator (or compact) if it is the limit of finite rank operators, $u = \lim_{n\to\infty} u_n$, with $u_n \in \mathscr{L}(E,F)^{\mathrm{fin}}$.*

We can give a more explicit characterization of the notion of a compact operator.

Lemma 8. *If $E = c(I)$, $u \in \mathscr{L}(E,F)$ defined by $u = (u_j)$ is compact if and only if $\lim_{j\to\infty} u_j = 0$. If $F = c(J)$, then this is equivalent to $\lim_{j\to\infty} \sup_{i\in I} \eta_{ij} = 0$.*

Proof. $u : E \to F$, reduce to the case where $\dim u(E) = 1$. □

Definition 13. *Let E, F and X be K-Banach spaces, then there is a K-Banach space $E\hat{\otimes}F$ such that we have a bijection between $\mathscr{L}(E\hat{\otimes}F, X)$ and bilinear maps $B(E\times F, X)$.*

The following lemma shows that we can extend scalars. If L/K, we have:

Lemma 9. *$L\hat{\otimes}c(J) \cong c_L(J)$, where $c_L(J)$ denotes the subspace of sequences of elements of L that go to 0.*

3.1.2 Fredholm determinant

Let R be a commutative ring, let L be a free R-module, and let $u : L \to L$ be an endomorphism of finite rank. Then, if M is a finite free module that contains $u(L)$, we can define $\det(1 - tu|_M)$, and this is independent of M.

Now let E be a K-Banach space, $u \in \mathscr{C}(E,E)$, and we would like to define $\det(1-tu)$. Let A be the valuation ring of K, and let $E_0 = \{x \in E \text{ such that } |x| \leqslant 1\}$. Assume that $|u| = 1$ which can be done after multiplying by a constant, and we have $u : E_0 \to E_0$. Let α be an ideal of A. Then, u induces a map of A/α-modules $u_\alpha : E_0/\alpha E_0 \to E_0/\alpha E_0$. Since $u \in \mathscr{C}(E,E)$, we have finitely many $i \in I$ such that $u_\alpha(e_i) \notin \alpha$, so we are in the situation described above, and we can define $\det(1 - tu_\alpha) \in A/\alpha[t]$.

Definition 14. *Given $u \in \mathscr{C}(E,E)$, we define its Fredholm determinant as $\varprojlim_\alpha \det(1 - u_\alpha t)$.*

Remark 10. *Perhaps the equivalent definition given in [2], pg 12, is more natural: For S any finite subset of I, define $c_S = \sum_{\sigma:S\to S} \operatorname{sgn}(\sigma) \prod_{i\in S} a_{i\sigma(i)}$, $c_n = (-1)^n \sum_{|S|=n} c_S$ and $\det(1-ut) = \sum_n c_n X^n$. In both cases is the limit over finite subspaces of the characteristic polynomial of u restricted to the projection to the finite subspaces.*

Theorem 3. *(Properties of Fredholm determinant) Let $E = c(I)$, and let $u \in \mathscr{C}(E,E)$. Then,*

1. $\det(1 - tu) = \sum_{m=0}^{\infty} c_m t^m$
2. $\det(1 - tu)$ *is a p-adic entire function (that is, $\in \mathscr{O}(\mathbb{A}^{1,an})$).*
3. *If $u_n \to u$, $u_n \in \mathscr{C}(E,E)$, then $\lim_{n\to\infty} \det(1 - tu_n) = \det(1 - tu)$.*

Corollary 1.

- $\det(1 - tu)(1 - tv) = \det(1 - tu)\det(1 - tv)$ *for all $u, v \in \mathscr{C}(E,E)$.*
- *For $u \in \mathscr{C}(E,F)$, $v \in \mathscr{C}(F,E)$, then $\det(1 - t(u\circ v)) = \det(1 - t(v\circ u))$.*

Definition 15. *We define the trace of $u \in \mathscr{C}(E,E)$ as the coefficient of t in $\det(1 - tu)$.*

Remark 11. *Given a p-adic entire function $p(T) = \sum_n a_n T^n \in \mathscr{O}(\mathbb{A}^{1,an})$ (that is, $\lim_{n\to\infty} |a_n| R^n = 0$ for every $R > 0$), we can consider the restriction to the fist of radius p^n, $\mathscr{O}(\mathbb{A}^{1,an}) \to K\langle p^{-n}T\rangle$. Then, Weierstrass preparation applies, and we have a factorization $p(T) = Q(T)U(T)$, with U a unit, and so all the roots of $p(T)$ in $B(0,p^{-n})$ are roots of Q and there are finitely many. In particular, this shows that we can order decreasingly the eigenvalues (or slopes) of $\det(1 - tu)$. The operator u has thus a discrete spectrum. Observe that the condition that we strongly use to obtain this discrete spectrum is that u is completely continuous.*

3.1.3 Riesz theory

Theorem 4. *(Riesz Theory, c.f. [9], Proposition 12) Let $P(t) = \det(1 - tu)$ be the Fredholm determinant of a completely continuous operator u. Let a be a zero of order h of $P(t)$, that is, $\frac{d^k P}{dt^k}(a) = 0$ for $k \leqslant h$ and $\frac{d^{h+1} P}{dt^{h+1}}(a) \neq 0$. Then, we have the following decomposition*

$$E = N(a) \oplus F(a)$$

satisfying

1. *$N(a)$ and $F(a)$ are closed subspaces, stables for u.*
2. *$1 - au$ is nilpotent in $N(a)$.*
3. *$N(a)$ has finite dimension equal to h.*
4. *$1 - au$ is invertible in $F(a)$.*

Theorem 5. *(c.f. [2], Theorem 3.3) Let $P(t) = \det(1 - tu)$ be the Fredholm determinant of a completely continuous operator u, and let $S = 1 + \ldots$, $Q = 1 + \ldots$, with $Q(t)$ a polynomial of degree n, and $S(t)$ a everywhere convergent power series that is a unit. Then there is a unique decomposition of u-invariant closed subspaces $E = N \oplus F$ such that $Q^*(u)$ is zero on N and invertible on F. Moreover, N has dimension n and $Q(t)$ is the characteristic power series of $u|_N$.*

Remark 12. *In [2] some of the results are stated in more generality, and we use them in the following sections of these notes. In particular, some of the generalizations are:*

- *Instead of a K-Banach space E, we consider a Banach A-module R over a Noetherian K-Banach algebra A. Then the Fredholm series is a function on $\mathrm{Sp}(R) \times \mathbb{A}^1$.*
- *There is a slight generalization of potentially ON, called property (Pr), which is the equivalent of projective modules in this setting, and is equivalent to being a direct summand of a potentially ON module.*

3.2 p-adic modular forms

Here we follow [8]. We want to find an operator U acting on a certain K-Banach space of p-adic modular forms $M_k^\dagger(N)$, that contains the classical ones, and such that U acts on $M_k^\dagger(N)$ completely continuously.

Let $X_1(N)$ be the modular curve of level $\Gamma_1(N)$, and X_1^{ord} the ordinary locus, which is an affinoid subdomain of X^{an}. One could think of defining p-adic modular forms as $H^0(X^{ord}, \omega^{\otimes k})$: According to [3] this is the first natural definition because it agrees with Serre's approach using q-expansions, that is, A_p Hasse invariant is $\equiv 1 \mod p$, so $A^{p^n - 1}$ converges to A^{-1}, and so the Hasse invariant is invertible. Defining modular forms as global sections over the ordinary locus is therefore a geometric parallel of what Serre does.

However, this is not entirely satisfactory, as U_p has a continuous spectrum. If V_p is a lift of the Frobenius operator, it can be shown that $V_p U_p = 1$. Then, if $f = \sum_n a_n q^n \in \ker U_p$, we let $\lambda \in K$ with $|\lambda| < 1$, and we can define a new p-adic modular form $f_\lambda = \sum_{n \geqslant 0} \lambda^n V_p^n f$, and using $U_p f = 0$ and $U_p V_p = 1$ we get

$$U_p g = U_p f + \sum_{n \geqslant 1} \lambda^n (U_p V_p) V_p^{n-1} f = \lambda \sum_{n \geqslant 0} \lambda^n V_p^{n-1} f = \lambda g$$

So for every λ we get an eigenform of value λ. This gives a continuous spectrum, and so it doesn't provide us with a good spectral decomposition like the one that we have over $\mathbb{C}$. To sort this out, we introduce overconvergent eigenforms.

Remark 13. *We have a map can : $X^{ord} \to Y = X(\Gamma_1(N) \cap \Gamma_0(p))$ given by $(E, i) \mapsto (E, i, H)$, where H is the canonical subgroup. By the Theorem of Katz and Lubin that we saw last time, we can extend can to $X^{\leqslant v}$ for $v < \frac{p}{p+1}$, where $X^{\leqslant v}$ is the overconvergent locus of elliptic curves with Hasse invariant $v(A(E, \omega)) \leqslant v$.*

Definition 16. *Let $v < \frac{p}{p+1}$. Then, we define the space of overconvergent modular forms of slope v and weight k as*

$$M_k^\dagger(v) = H^0(X^{\leqslant v}, \underline{\omega}^{\otimes k})$$

We want to define the operator U_p. We treat Hecke operators as correspondences between the modular curves. We consider the auxiliary curve $Z = X(\Gamma_1(N) \cap \Gamma_0(p) \cap \Gamma^0(p))$, which parametrizes tuples (E, i, H, C), with E elliptic curve, $i : \mathbb{Z}/N\mathbb{Z} \hookrightarrow E[N]$ level structure, H, C finite flat subgroup schemes of order p with $C \cap H = 1$. Then, we have the following diagram:

$$\begin{array}{ccc} & Z = X(\Gamma_1(N) \cap \Gamma_0(p) \cap \Gamma^0(p)) & \\ {\scriptstyle p_1}\swarrow & & \searrow{\scriptstyle p_2} \\ Y = X(\Gamma_1(N) \cap \Gamma_0(p)) & & Y = X(\Gamma_1(N) \cap \Gamma_0(p)) \end{array}$$

With $p_1(E, H, C) = (E, H)$ and $p_2(E, H, C) = (E/C, H \mod C)$. We have $p_1^*\underline{\omega}^k \cong p_2^*\underline{\omega}^k$, and we define the U_p operator as follows.

Definition 17. *The U_p operator on $Y_{\mathbb{Q}_p} = X(\Gamma_1(N) \cap \Gamma_0(N))_{\mathbb{Q}_p}$ is the map*

$$H^0(Y_{\mathbb{Q}_p}, \underline{\omega}^k) \xrightarrow{p_2^*} H^0(Z_{\mathbb{Q}_p}, p_2^*\underline{\omega}^k) \xrightarrow{\cong} H^0(Z_{\mathbb{Q}_p}, p_1^*\underline{\omega}^k) \xrightarrow{(p_1)_*} H^0(Y_{\mathbb{Q}_p}, (p_1)_*p_1^*\underline{\omega}^k) \xrightarrow{1/p\,Tr} H^0(Y_{\mathbb{Q}_p}, \underline{\omega}^k)$$

Where the trace is well defined because p_1 is finite flat, and in particular locally free.

Remark 14. *What this is doing geometrically is averaging over all subgroups of order p that are not the canonical subgroup. This agrees with the geometric way to see the U_p operator as a map of divisors of $X_1(N)$ (for instance over $\mathbb{C}$). Moreover, it can be checked that this definition of the U_p-operator agrees with the classical definition on q-expansions, by evaluating at the Tate curve: Observe that* $\mathrm{Tate}(q)/\zeta_p^i q^{1/p} \cong \mathbb{G}_m/\langle \zeta_p^i q^{1/p} \rangle = \mathrm{Tate}(\zeta_p^j q^{1/p})$

$$U_p f(\mathit{Tate}(q), \mu_p) = \frac{1}{p}\sum_{j=0}^{p-1}(f(\mathrm{Tate}(\zeta_p^j q^{1/p}))) = \sum_{j=0}^{p-1}(\sum_{n=0}^{\infty} a_n(\zeta_p^j q^{1/p})^n) = \sum_{n=0}^{\infty} a_n \sum_{j=0}^{p-1} q^{n/p}\zeta_p^{in}$$

Now observe that $\sum_{j=0}^{p-1} p^{n/p}\zeta_p^{in} = \begin{cases} p & \text{if } n \mid p \\ 0 & \text{otherwise} \end{cases}$ *. Therefore, we get*

$$U_p f(\mathit{Tate}(q), \mu_p) = \sum_{n \geqslant 0} a_{np} q^n$$

Now, restrict the diagram to the neighborhoods of the ordinary locus $Y^{\leqslant v}$, and consider for simplicity $N = 1$, so no level N structure. We have the following diagram

$$\begin{array}{ccc} & p_1^{-1}(Y^{\leqslant v}) & \\ {\scriptstyle p_1}\swarrow & & \searrow{\scriptstyle p_2} \\ Y^{\leqslant v} & & Y^{\leqslant v/p} \end{array}$$

Observe that the fact that the image of p_2 lands in $Y^{\leqslant v/p}$ is a consequence of Theorem 2. Then the U_p operator can be defined as $U_p : H^0(X^{\leqslant v/p}, \underline{\omega}^k) \to H^0(X^{\leqslant v}, \underline{\omega}^k)$. By abuse of notation we also denote U_p the composition $H^0(X^{\leqslant v}, \underline{\omega}^k) \to H^0(X^{\leqslant v}, \underline{\omega}^k) \xrightarrow[U_p]{} H^0(X^{\leqslant v}, \underline{\omega}^k)$, where the first map is just restriction. Now we can show:

Theorem 6. *Let $0 < v < \frac{p}{p+1}$. Then the operator $U_p : H^0(X^{\leqslant v}, \underline{\omega}^k) \to H^0(X^{\leqslant v}, \underline{\omega}^k)$ is completely continuous.*

Proof. Observe that if $u : E \to F$ and $v : F \to G$ are continuous linear operators, and v or u are completely continuous, then so is $v \circ u$. This is an immediate consequence of the notion of completely continuous as a limit of finite rank operators (and finite rank operators satisfying the property of composition above). Therefore, as U_p factors through the restriction map $H^0(X^{\leqslant v}, \underline{\omega}^k) \to H^0(X^{\leqslant v}, \underline{\omega}^k)$, it is enough to prove that this is compact. Observe that this can be rewritten as

$$H^0(X^{\leqslant v}, \underline{\omega}^k) \to \mathscr{O}(X^{\leqslant v/p})\hat{\otimes}_{\mathscr{O}(X^{\leqslant v}}H^0(X^{\leqslant v}, \underline{\omega}^k)$$

Therefore it is enough to show that the restriction map on affinoid algebras is compact. This is an exercise in Newton's notes!. □

4 Coleman's Classicity theorem

We follow [4]. Let K be a complete discrete valued field, subfield of $\mathbb{C}_p$, $N \geqslant 5$ and $p \nmid N$ a prime, and $k \in \mathbb{Z}$. Let M_k denote the space of k-overconvergent modular forms of level $\Gamma_1(N)$ with coefficients in K. If $M_{k,cl}$ denotes the space of classical modular forms of weight k and level $\Gamma_1(N) \cap \Gamma_0(p)$, we have $M_k \supseteq M_{k,cl}$. Let's denote by U the U_p-operator.

Definition 18. *Given a $K[U]$-module M, and $\alpha \in \mathbb{Q}$, the slope α part of M, denoted M_α is the subspace of $F \in M$ such that $\exists f(x) \in K[X]$, all of whose roots have valuation α, such that $f(U)F = 0$.*

For this definition, the situation to keep in mind is the case in which F is a generalized eigenvector for α. The goal of today is to treat the following theorem:

Theorem 7. *(Coleman) For $k \geqslant 0$ and $\alpha < k+1$, the inclusion $(M_{k+2,cl})_\alpha \subseteq (M_{k+2})_\alpha$ is an equality.*

Remark 15. *The philosophy of Coleman's theory can be seen as studying modular forms by putting them into "families". In some sense, this allows us to use more structure to study them, as we have more points. Then, it is important to be able to distinguish which elements of the family are our classical objects of study, which is what Coleman's classicity theorem does.*

The rough idea of the proof consists of two steps. We identify a sub-$K[U]$-module $Z \subseteq M_{k+2}$ such that, if $\alpha < k+1$,

- $Z_\alpha = 0$
- $\dim(M_{k+2}/Z)_\alpha) = \dim(M_{k+2,cl})_\alpha$

To do this, we use the de Rham complex over neighborhoods of the ordinary locus where the canonical subgroup lives, to construct a U-equivariant map

$$\Theta^{k+1} : M_{-k} \to M_{k+2}$$

that on q-expansions is given by $f(q) \mapsto \left(q\frac{d}{dq}\right)^{k+1} f(q)$. This operator does not preserve modularity, but it preserves modularity $\mod p$.

Then, by the boundedness of coefficients in M_{-k}, $Z = \Theta^{k+1}(M_{-k})$ satisfies $Z_\alpha = 0$ if $\alpha < k+1$. Then, we prove that coker(Θ^{k+1}) is (almost) the de Rham cohomology of $X_1(N)$, hence its dimension can be computed in terms of classical modular forms, via Eichler-Shimura correspondence. This is what allows us to get the dimension comparison $\dim(M_{k+2}/Z)_\alpha) = \dim(M_{k+2,cl})_\alpha$.

4.1 A quick première on de Rham cohomology

Let A be a commutative ring, B a commutative A-algebra, $\Omega^1_{B/A}$ Kahler differentials, and $\Omega^q_{B/A} = \bigwedge^q_A \Omega^1_{B/A}$. We get a complex

$$B \to \Omega^1_{B/A} \to \Omega^2_{B/A} \to \dots$$

with differentials given by $b_0 db_1 \wedge \dots \wedge db_q \mapsto db_0 \wedge \dots db_q$. This complex sheafiffies and, for any $f : X \to S$ morphism of schemes, we get $\Omega^\bullet_{X/S}$ a complex of quasi-coherent sheaves of $\mathscr{O}_S$-modules.

Example 4. *1. $\mathbb{A}^1_K$, $K[t] \to K[t]dt$ given by $f(t) \mapsto f'(t)dt$.*

2. $\mathbb{P}^1_K$ locally looks like $\mathbb{A}^1$, but in global sections it is just the zero map.

3. $\mathbb{G}_m$, $K[t,t^{-1}] \to K[t,t^{-1}]dt$ is given by $f(t) \mapsto f'(t)dt$, but observe that $\frac{dt}{t}$ is not in the image!

Definition 19. *The de Rham cohomology is*

$$H^i_{dR}(X/S) = R^i\Gamma(X, \Omega^\bullet_{X/S})$$

We can also define a sheaffified (relative) version

$$R^i f_* \Omega^\bullet_{X/S} \in \mathrm{QCoh}(\mathscr{O}_S)$$

Recall that the nice thing about having different cohomology theories is that each of them has its own features. The de Rham cohomology comes with a filtration, which is given by

$$\Omega^{\geqslant q}_{X/S} : 0 \to 0 \to \cdots \to \Omega^q_{X/S} \to \Omega^{q+1}_{X/S} \to \dots$$

This filtration is called the Hodge filtration on $H^n_{dR}(X/S)$,

$$\mathrm{Fil}^q H^n_{dR}(X/S) = \mathrm{Im}(R^n\Gamma(X, \Omega^{\geqslant q}) \to R^n\Gamma(X, \Omega^\bullet_{X/S}))$$

A filtered complex gives a spectral sequence

$$E_1^{p,q} := H^q(X, \Omega^p_{X/S}) \implies H^{p+q}_{dR}(X/S)$$

Example 5. *Let X/K be a smooth projective curve. Then, the filtration is*

$$\begin{array}{ccccc} 0 & \longrightarrow & \Omega^1_{X/K} & = & \Omega^{\geqslant 1}_{X/K} \\ & & & & \cap| \\ \mathscr{O}_X & \longrightarrow & \Omega^1_{X/K} & = & \Omega^\bullet_{X/K} \end{array}$$

and the quotient is $\mathscr{O}_X$. So taking cohomology we get

$$H^0(X, \Omega^1_X) = H^1(X, \Omega^1_X[-1]) \to H^1_{dR}(X) \to H^1(X, \mathscr{O}_X) \cong H^0(X, \mathscr{O}_X)^\vee$$

where the last isomorphism is by Serre duality. It can be shown that for $\mathrm{char}K = 0$ this sequence is exact.

Now we consider a variant of de Rham cohomology: We often want to work on a compactified X over some scheme Y. Say Y is a smooth separated variety over K, $Y \to X$ open immersion, and $D = X \backslash Y$ is a normal crossings divisor (which roughly means that étale locally is cut out by a regular sequence).

Example 6. $\mathbb{G}_m \subseteq \mathbb{P}^1$ *has $D = \mathbb{P}^1 \backslash \mathbb{G}_m = \{0, \infty\}$ cut out by $\{[x : y] \in \mathbb{P}^1$ such that $xy = 0\}$. This is the nicest type of singularity that we can have.*

Then, in this case, we define

$$\Omega^q_{X/K}(\log D)(U) = \{u \in \Omega^q_{Y/K} \text{ such that } \omega, d\omega \text{ have poles of order } \leqslant 1 \text{ on } D\}$$

For example, $\frac{dt}{t} \in \Omega^1_{\mathbb{P}^1/K}(\log D)$.

A fact is that $\Omega^\bullet_{X/K}(\log D)$ is a complex on X computing the de Rham cohomology of $H^i_{dR}(Y/K)$. this is what we do if we want/have to work over some compactification, like in the case of modular curves.

Now let $f : X \to S$ be a proper smooth map of separated K-schemes, with S smooth. We define a filtration on $\Omega^\bullet_{X/K}$ by

$$F^p\Omega^q_{X/K} = \mathrm{Im}(f^*\Omega^p_S \otimes_{\mathscr{O}_X} \Omega^{p-q}_X \to \Omega^q_X)$$

Then, we have

$$0 \to F^1/F^2 \to F^0/F^2 \to F^0/F^1 \to 0$$

And taking cohomology, the boundary map gives the *Gauss-Manin connection*

$$\nabla : R\Gamma f_*\Omega^0_{X/S} \to \Omega^1_S \otimes R^p f_*\Omega^0_{X/S}$$

It is a K-linear map satisfying $\nabla(fs) = df \otimes s + f\nabla(s)$, for $s \in R^p f_*\Omega^0_{X/S}(U)$, $f \in \mathscr{O}(U)$. We remark that if $\dim S = 1$, the same can be done with log poles.

4.2 Coleman's theorem

We first set up the setting: Let X be the smooth proper modular curve of level $\Gamma_1(N)/\mathscr{O}_K$, and $C \subset X$ be the cuspidal divisor. Let $f : E \to X$ be the universal generalized elliptic curve and let $\widetilde{C} = f^{-1}(C)$.

Now we have $\Omega^1_{E/X}(\log \widetilde{C}) := \Omega^1_E(\log \widetilde{C})/f^*\Omega^1_X(\log C)$, and $\omega = f_*\Omega^1_{E/X}(\log \widetilde{C})$ the usual invertible sheaf on X. We also have $E_{p-1} \in H^0(X, \omega^{\otimes p-1})$ the Eisenstein series lifting the Hasse invariant, and for $r \in \mathbb{Q}$,

$$X_r := \{x \in X^{an} \text{ such that } |E_{p-1}(x)| > r\}$$

Moreover, we denote $W_1 = X_{p^{-p/p+1}}$, where we have canonical subgroup, and $W_2 = X_{p^{\frac{-1}{p+1}}}$ which is the space such that, if we mod out by the canonical subgroup we still land in W_1. We have $X^{\text{ord}} \subseteq W_2 \subseteq W_1$.

For $k \in \mathbb{Z}$, set $M_k := \omega^k(W_1)$. For $k \geqslant 0$, let

$$\mathscr{H}_k := \text{Sym}^k R^1 f_* \Omega^\bullet_{E/X}(\log \widetilde{C})$$

Let $E_i = E \times W_i$, for $i = 1, 2$. Then we have maps

$$\begin{array}{ccc} E_2 & \xrightarrow{\Phi} & E_1 \\ \downarrow f & & \downarrow f \\ W_2 & \xrightarrow{\phi} & W_1 \end{array}$$

induced by $\pi : E_2 \to E_2/\mathscr{K}$ universal isogeny, where $\mathscr{K}$ is the family of canonical subgroups, and ϕ is defined by $\phi^* E_1 \cong E_2/\mathscr{K}$ (that is, we base change E_1 to W_2 and that is $\phi^* E_1$, which is isomorphic to $E_2/\mathscr{K}$. Then Φ is the composition of the projection $E_2 \to E_2/\mathscr{K}$ and the projection of the fibred product $\phi^* E_1 \to E_1$. An important fact shown by Katz is that ϕ and Φ are finite and étale of degree p.

If $x \in W_1$, corresponds to an elliptic curve E_x and canonical subgroup H_x, then the pre-images of E_x under ϕ are E_x/H_i, where $H_1, \dots H_p$ are the subgroups of order p of E_x that are not the canonical subgroup. We define operators V_k and F_k (Verschiebung and Frobenius) as follows:

$$\begin{array}{ccc} \mathscr{H}_k(W_2) & \xrightarrow{\breve{\pi}^*} & \text{Sym} R^1 f_* \Omega^1_{E_1^{\phi^*}/W_2}(W_2) \\ & \searrow^{V_k} & \cap \\ & & \phi^* \mathscr{H}_k(W_2) \\ & & \downarrow Tr_\phi \\ & & \mathscr{H}_k(W_1) \end{array}$$

$$\begin{array}{ccc} \mathscr{H}_k(W_1) & \xrightarrow{h \mapsto h^{\phi^*}} & \phi^* \mathscr{H}_k(W_2) \\ & \searrow^{F_k} & \downarrow \\ & & \text{Sym}^k R^1 f_* \Omega_{E_1^{\phi^*}/W_2}(W_2) \\ & & \downarrow \pi^* \\ & & \mathscr{H}_k(W_2) \end{array}$$

We have, for $h \in \mathscr{H}_j(W_2)$, $g \in \mathscr{H}_k(W_1)$, $V_{k+j}(hF(g)) = p^k V_j(h) g$, as $\pi^* \circ \breve{\pi}$ is just multiplication by p.

Now, Hodge filtration gives an exact sequence

$$0 \to \omega \to \mathscr{H}_1 \to \omega^{-1} \to 0$$

Which induces a filtration on all symmetric powers

$$\mathscr{H}_k = \text{Fil}^0 \mathscr{H}_k \supseteq \cdots \supseteq \text{Fil}^{k+1} \mathscr{H}_k = 0$$

with graded pieces $\text{Fil}^{k-i} \mathscr{H}_k / \text{Fil}^{k+1-i} \mathscr{H}_k \cong \omega^{k-2i}$.

Remark 16. *Observe that in the last graded step of the filtration we get the space of modular forms that we want, but we also get all the other weights for all the other graded pieces. In particular, we have*

$$M_j = \begin{cases} Fil^k \mathscr{H}_k(W_1) & j \geqslant 0 \\ Fil^0 \mathscr{H}_{-j}/Fil^1 \mathscr{H}_{-j}(W_1) & j < 0 \end{cases}$$

Definition 20. *We define the U-operator on M_j by*

$$U = \begin{cases} \frac{1}{p} V_j \circ Res_{W_2}^{W_1} & \text{if } j \geqslant 0 \\ p^{j-1} V_{-j} \circ Res_{W_2}^{W_1} & \text{if } j < 0 \end{cases}$$

Remark 17. *The same construction works with $\Omega^1_{X_1(N)} \otimes \mathscr{H}_k$ instead of $\mathscr{H}_k$*

Proposition 5. *Let $k \geqslant 0$. Then the map induced by $V_k \circ Res_{W_2}^{W_1}$ on*

$$Fil^{k-i} \mathscr{H}_k / Fil^{k+1-i} \mathscr{H}_k \cong M_{k-2i}$$

is $p^{i+1}U$, for $0 \leqslant i \leqslant k$.

Now we consider the Gauss-Manin connection $\nabla : \mathscr{H}_k \to \Omega^1_X(\log C) \otimes \mathscr{H}_k$, and we can check that it commutes with V_k and F_k. In particular, for $k = 1$, we get a map $\omega \subseteq \mathscr{H}_1 \xrightarrow{\nabla} \mathscr{H}_1 \otimes \Omega^1_X(\log C) \to \omega^{-1} \otimes \Omega^1_X(\log C)$, and then tensoring again with ω, we get

$$\kappa : \omega^2 \to \Omega^1_X(\log C)$$

which is the Kodaira-Spencer isomorphism. Moreover, this can be extended to

$$\kappa_k : \omega^{k+2} \to \Omega^1_X(\log C) \otimes \mathscr{H}_k$$

defined by $\alpha \otimes \beta \otimes \gamma \mapsto \kappa(\alpha \otimes \beta) \otimes \gamma$, if $\gamma \in \omega^k$.

Lemma 10. *The map*

$$Fil^1 \mathscr{H}_k \xrightarrow{\nabla} \frac{\Omega^1_X(\log C) \otimes \mathscr{H}_k}{\Omega^1_X(\log C) \otimes Fil^k \mathscr{H}_k}$$

is an isomorphism of sheaves of K-vector spaces.

Proof. (Sketch) ∇ in general does not respect the filtration, but it is good enough, namely it sends Fil^i to Fil^{i-1}, what is called Griffiths transversality:

$$\nabla(\mathrm{Fil}^i \mathscr{H}_k) \subseteq \Omega^1_X(\log C) \otimes \mathrm{Fil}^{i-1} \mathscr{H}_k$$

This implies that we have

$$\mathrm{gr}\nabla : \bigoplus_{i=0}^{k-1} \omega^{k-2i} = \omega^2 \oplus \bigoplus_{i=0}^{k-1} \omega^{k-2(i+1)} \to \Omega^1_X(\log C) \otimes \bigoplus_{i=0}^{k-1} \omega^{k-2(i+1)}$$

Observe that tensoring with $KS \otimes id$ gives an isomorphism, so it has to be an isomorphism. □

As a corollary, we get an interesting map

$$M_{-k} \to M_{k+2}$$

defined as follows: Take $\omega \in M_{-k} = \mathscr{H}_k(\mathrm{Fil}^1 \mathscr{H}_k(W_1))$ and lift it to $\hat{\omega} \in \mathscr{H}_k(W_1)$. Let $s \in \mathrm{Fil}^1 \mathscr{H}_k(W_1)$ such that $\nabla\hat{\omega} = \nabla s \mod \Omega^1_X(\log C) \otimes \mathrm{Fil}^k \mathscr{H}_k(W_1) = M_{k+2}$. Then the map is $\omega \mapsto \nabla(\hat{\omega} - s)$. Then, Katz shows that this map on q-expansions is $f(q) \mapsto \frac{(-1)^k}{k!} \left(q \frac{d}{dq}\right)^{k+1} f(q)$. Set $\Theta^{k+1} = k!(-1)^k \nabla(\hat{\omega} - s)$.

Now let's denote $SS \subset X_1(N) \mod p$ the set of supersingular points, and $\widetilde{SS}$ the set of lifts.

Lemma 11 (Baldassari-Chiarelotto)**.** *Let $H(k)(W_i) := (\Omega^1_X(\log C) \otimes \mathscr{H}_k)(W_i)/\nabla \mathscr{H}_k(W_i)$, then we have*

$$H(k)(W_i) \cong H^1(X_1(N), \Omega^\bullet(H_k)(\log \widetilde{SS}))$$

And, in particular, $H(k)(W_i)$ are finite dimensional, and they don't depend on i.

Theorem 8 ([4], 5.4)**.** *$M_{k+2}/\Theta^{k+1}M_{-k} \cong H(k)$, and the following diagram commutes:*

$$\begin{array}{ccc} M_{k+2}/\Theta^{k+1}M_{-k} & \xrightarrow{U} & M_{k+2}/\theta^{k+1}M_{-k} \\ \downarrow & & \downarrow \\ H(k) & \xrightarrow{Ver} & H(k) \end{array}$$

Corollary 2. *Suppose that we have $\alpha \in \mathbb{Q}$, with $\alpha < k+1$. Then, the map $(M_{k+2,cl})_\alpha \to H(k)_\alpha$ is injective.*

Proof. (of the corollary) By the theorem, it suffices to show that $(\Theta^{k+1}M_{-k})_\alpha = 0$. Take $F \in M_{k+2}$ a U-eigenvector with eigenvalue λ and $v_p(\lambda) < k+1$. Assume $F = \theta^{k+1}G$ for some $G \in M_{-k}$. Now we look at q-expansions, $F(q) = \sum a_n q^n$ and $G(q) = \sum b_n q^n$, we have $a_{pn} = \lambda a_n$, and $a_n = n^{k+1}b_n$, $\forall n \geqslant 0$. This implies $b_{p^r} = (\lambda_{p^{-(k+1)}})^r a_1 \to \infty$ as $r \to 0$, which is a contradiction. □

Now to finish the proof of Coleman's classicity, we argue by dimension: We show that $\dim(M_{k+2,cl})_\alpha = \dim H(k)_\alpha$. On the left hand, we just count modular forms of weight $k+2$ and level $\Gamma_1(N) \cap \Gamma_0(N)$ of slope α. On the right hand, we use Eichler-Shimura and some residue computation.

5 Overconvergent modular forms of p-adic weight (geometrically)

From now on, we introduce modular forms of p-adic weight. There are 3 approaches to defining such modular forms. In historical order, they're:

- Using q-expansions (the one Coleman originally did, we will treat next time)
- Using modular sheaves of p-adic weight (the approach that we will follow today)
- Using perfectoid towers of modular curves and the Hodge-Tate map.

Although historically it would make sense to start with the q-expansion approach, we start with the geometric approach for logistic reasons.

Remark 18. *The main advantage of the geometric approach explained today, as opposite to the classical one introduced by Coleman, is that it does not rely on Eisenstein Series, and therefore it is possible to generalize it to other groups.*

Recall that given p a prime, $p \geqslant 5$, $N \geqslant 5$ and $p \nmid N$, $X = X_1(N)^{an} \to \mathrm{Spa}(\mathbb{Q}_p, \mathbb{Z}_p)$. We have $\pi : E \to X$ the universal generalized elliptic curve. For $r \geqslant 0$, we denote $X_r = \{x \in X \text{ such that } |E_{p-1}^{p^{r+1}}|_x \geqslant |p|_x\}$. If x is a closed point, $|E_{p-1}(x,\omega)| \geqslant \frac{1}{p^{\frac{1}{p^{r+1}}}}$. As we are using adic spaces, non-closed points could be more complicated.

If $k \in \mathbb{Z}$, denote as usual $M_k(r) = H^0(X_r, \underline{\omega}^{\otimes k})$, $M_k^\dagger = \lim_r M_k(r)$. It has an action of the Hecke operators T_l, for $l \nmid N$ and U_p. Recall that we work with overconvergent modular forms because then U_p is completely continuous, so the infinite dimensional spaces can be cut out by slopes. Now we want to introduce p-adic weights. It is natural to ask:

Question: Why should p-adic weights exist?

What shows that p-adic weights appear naturally is the following theorem by Serre:

Theorem 9. *Let $\{f_n = f_n(q)\}$, for $f_n \in \mathbb{C}_p[[q]]$ be a sequence of classical modular forms of weights k_n. Suppose that $\{f_n\}$ converges uniformly to $f \in \mathbb{C}_p[[q]]$. Then, the sequence $\{k_n\}$ converges to some $k \in \mathbb{Z}_p/(p-1)\mathbb{Z}_p \times \mathbb{Z}_p$. We say that f is a p-adic modular form of weight k.*

Remark 19.

1. *The topology on $\mathbb{C}_p[[q]]$. We define a valuation*

$$\begin{aligned} v :& \mathbb{C}_p[[q]] \to \mathbb{Q} \cup \{\pm\infty\} \\ & \sum a_n q^n \mapsto \inf_n v(a_n) \end{aligned}$$

   If $\{f_n\}_n, f \in \mathbb{C}_p[[q]]$, $f_n \to f$ uniformly $\iff$ $v(f - f_n) \to \infty$.

2. *What is the group $\mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}_p$? It contains $\mathbb{Z}$ embedded diagonally:*

$$\begin{aligned} \mathbb{Z} \subseteq & \mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}_p \to Hom_{\mathrm{cont}}(\mathbb{Z}_p^\times, \mathbb{Z}_p^\times) \\ & (a,b) \mapsto [t = (\bar{t}, \langle t \rangle) \mapsto \bar{t}^a \langle t \rangle^b] \end{aligned}$$

   Observe also that we have $Hom_{\mathrm{cont}}(\mathbb{Z}_p^\times, \mathbb{Z}_p^\times) \cong Hom_{\mathrm{cont}}(\mathbb{Z}_p^\times, \mathbb{Q}_p^\times) \cong Hom_{\mathrm{cont,alg}}(\Lambda, \mathbb{Q}_p)$, with $\Lambda = \mathbb{Z}_p[[\mathbb{Z}_p^\times]]$. We write this way because it gives something geometric, and we can view $\mathbb{Z}/(p-1)\mathbb{Z} \times \mathbb{Z}_p$, as $W(\mathbb{Q}_p)$, the $\mathbb{Q}_p$-points of the weight space, associated to $\mathrm{Spf}(\Lambda)$. *W is a disjoint union of $(p-1)$ open disks, $\Lambda \cong (\mathbb{Z}_p[[T]])^{p-1}$.*

3. *If R is any p-adically complete $\mathbb{Q}_p$-algebra, $W(R) = Hom_{\mathrm{cont}}(\Lambda, R) \cong Hom_{\mathrm{cont}}(\mathbb{Z}_p^\times, R^\times)$. Again we have $\mathbb{Z} \subseteq W(R)$. Let's give another example of weights: Suppose that $U = \mathrm{Spa}(A, A^+) \hookrightarrow W$. We have $\kappa_U^{\mathrm{univ}} : \Lambda \to \mathscr{O}_W^+(W) \xrightarrow{\mathrm{res}} A^+$ universal weight associated to the open U. It can also be seen as $\mathbb{Z}_p^\times \to \Lambda^\times$, $t \mapsto t$.*

Now recall that we wish to do the following: For $\kappa \in W(R)$, R p-adically complete $\mathbb{Q}_p$-algebra, we wish to find $r \geqslant 0$ and a sheaf $\underline{\omega}^\kappa$ on $X_{r,R}$ such that

1. If $k \in \mathbb{Z} \subseteq W(\mathbb{Q}_p)$, we have $\underline{\omega}^\kappa \cong (\underline{\omega}_E|_{X_r})^{\otimes k}$.
2. $M_k(r) \cong H^0(X_{r,R}, \underline{\omega}^k)$.

In other words, we want to p-adically interpolate the family of sheaves $\{(\underline{\omega}_E|_{X_r})^{\otimes k}\}_{k \in \mathbb{Z}}$.

5.1 Vector bundles with marked sections

Let S be an adic space, $(\mathscr{O}_S, \mathscr{O}_S^+)$. Let $\mathscr{E}$ be a locally free ${\mathscr{O}_{\mathscr{S}}}^+$-module of rank d. Let $J \subseteq \mathscr{O}_S^+$ be an invertible ideal (i.e. J is coherent and locally generated by one element, which is not a zero divisor).

Definition 21. *Let $\overline{\mathscr{E}} = \mathscr{E}/J\mathscr{E}$, $\overline{\mathscr{O}}_S^+ = \mathscr{O}_S^+/J\mathscr{O}_S^+$. Suppose that $e_1, \dots, e_d \in H^0(S, \overline{\mathscr{E}})$ is a basis as a $H^0(S, \overline{\mathscr{O}_S^+})$-module. Then, the data $(\mathscr{E}, \{e_1, \dots, e_d\})$ is called a locally free sheaf with marked sections.*

Let $\mathscr{C}$ be the category whose objects are morpisms of adic spaces $t : T \to S$ such that $t^*(J)$ is invertible in $\mathscr{O}_T^+$.

Theorem 10. *1. The functor on $\mathscr{C}$ defined as*

$$(T \xrightarrow{t} S) \mapsto Hom_{\mathscr{O}_T^+}(t^*\mathscr{E}, \mathscr{O}_T^+)$$

is represented by an adic space $\mathbb{V}(\mathscr{E}) \to S$, called an adic vector bundle.

2. The functor on $\mathscr{C}$ defined as

$$(T \xrightarrow{t} S) \mapsto \{h \in Hom_{\mathscr{O}_T^+}(t^*\mathscr{E}, \mathscr{O}_T^+) \text{ such that } h(t^*(e_i)) = 1\}$$

is represented by an adic space $\mathbb{V}(\mathscr{E}, e_1, \dots e_d) \to S$, and we have

$$\begin{array}{ccc} \mathbb{V}(\mathscr{E}, e_1, \dots e_d) & \longrightarrow & \mathbb{V}(\mathscr{E}) \\ & \searrow \quad \swarrow & \\ & S & \end{array}$$

$\mathbb{V}(\mathscr{E}, e_1, \dots e_d)$ is called a vector bundle with marked sections (VBMS).

Let's denote $\mathscr{F} := \rho_* \mathscr{O}^+_{\mathbb{V}(\mathscr{E}, e_1, \dots e_d)}$, where $\rho : \mathbb{V}(\mathscr{E}, e_1, \dots e_d) \to S$.

Now, in the modular curves situation, there are such VBMS. Recall again that we have $E \xrightarrow{\pi} X = X_1(N)^{\text{an}} \supseteq X_r \supset X^{\text{ord}}$. Let $r \geqslant 0$, $X_r \ni x = (E_x, \alpha_x)$. There is a canonical subgroup $H_x \subseteq E_x[p]$. The canonical subgroups glue together and give $H/X_r \subseteq E[p]$. Let $H^D(X_r)$ be the Cartier dual of the canonical subgroup. In the following we will assume for simplicity that $H^D(X_r)$ has all its points, i.e $H^D(X_r) \cong \mathbb{Z}/p\mathbb{Z}$. In general this is only true after an extension to the Igusa tower.

Definition 22. *We define the Hodge-Tate map*

$$H^D(X_r) \to \underline{\omega}_{H/X_r}$$

as follows: A point $P \in H^D(X_r)$ corresponds to a morphism of group schemes over X_r, $P \in Hom_{\text{Gr}-\text{Sch}/X_r}(H, \mathbb{G}_{m,X_r})$. On themultiplicative group scheme we have a canonical differential dT/T, and we define

$$HT(P) := P^*(dT/T) \in \underline{\omega}_{H/X_r}$$

Now, we will use the Hodge-Tate map to produce a vector bundle with marked sections: Define $\omega^{\#} := \varphi^{-1}(HT(H^D) \otimes \mathscr{O}_{X_r}^+) \subseteq \underline{\omega}_E^+$, where φ is the natural map $\omega_E \to \omega_{H/X_r}$ φ is here a map between sheaves of differentials

Falta una pagina per picar!!

Apparently, the advantatge of this approach is that it doesn't depend on the Eisenstein series such as the classical approach does! Could be generalized to other settings.

6 Overconvergent modular forms of p-adic weight (classically)

6.1 Coleman families of modular forms

We will introduce Coleman's families of modular forms and explain the proof of the theorem below, following [5].

Theorem 11. *Let $\alpha \in \mathbb{Q}$, $\epsilon : (\mathbb{Z}/p\mathbb{Z})^\times \to \mathbb{C}^\times$, $\epsilon = \omega^i$, ω Teichmüller character. Then, there exists an integer $M = M(p, N, \epsilon, \alpha)$ such that, if $k > \alpha + 1$ and there exists a unique $F \in S_k(\Gamma_1(Np), \epsilon\omega^{-k})$ normalized eigenform of slope α, then for $k' \geqslant \alpha + 1$ such that $k \equiv k' \mod p^{M+n}$ there is a unique $F' \in S_{k'}(\Gamma_1(Np), \epsilon\omega^{-k'})$ of slope α, eigenform such that $F'(q) \equiv F(q) \mod p^{n+1}$.*

Observe that this gives a family of eigenforms varying in weight, on a neighborhood of radius M of the weight k.

Remark 20. *Observe that the statement of the theorem is saying that the dimension of the spaces of cusp forms doesn't vary with weight (dimension 1 in the case above), if we restrict the weights to a small enough disk of radius M. It is a natural question to wonder what should M be. Gouvea and Mazur conjectured that $M = 0$ if in the statement of the above theorem we impose $n \geqslant \alpha$. In the $\alpha = 0$ situation this is known by Hida, and in the general case it was disproved by Buzzard-Calegari, finding a counterexample for $p = 59$, $N = 1$. The problem is that for $p = 59$, $\Gamma_0(p)$ is not p-regular. A more realistic expectation would be that M is linear on α. This would be a consequence of the Halo conjecture.*

Let X_v denote the neighborhoods of the ordinary locus, and let $M_k(v)$ be the space of overconvergent modular forms, where U_p acts completely continuous. We denote $U_{p,k}$ if we want to specify the action of U_p on the weight k space.

Remark 21 (Twist of U_p)**.** *Suppose that $F \in M_{k-r}(v)$ is an overconvergent form such that $F^{-1} \in M_{r-k}(v)$ is overconvergent too. Then, we have an isomorphism*

$$\begin{aligned} M_k(v) &\to M_r(v) \\ h &\mapsto hF \end{aligned}$$

And we have $U_{p,r}\left(h\frac{F}{\sigma(F)}\right) = F^{-1}U_{p,k}(hF)$, where σ is the map $\sigma = \text{Frob}/p^k$, and $\text{Frob} : M_k(v) \to M_k(v/p)$. Observe that σ acts on q-expansions as $\sigma(F)(q) = F(q^p)$.

As a consequence of this result, we have:

Corollary 3. *The spectral theory for $U_{p,k}$ is the same as the spectral theory for the operator $U_{p,r}m_{F/\sigma F}$, where $m_{F/\sigma F}$ is just multiplication by $F/\sigma F$.*

Remark 22. *This is what is called a "link" in Buzzard's terminology:*

$$\begin{array}{ccc} M_r & \xrightarrow{F} & M_k \\ \downarrow{\scriptstyle U_{p,r}} & & \downarrow{\scriptstyle U_{p,k}} \\ M_r & \xrightarrow{\sigma(F)} & M_k \end{array}$$

We have an explicit example of such a form F: Recall that we have a family of Eisenstein series

$$G_\kappa(q) = \frac{\zeta_p^*(\kappa)}{2} + \sum_{n \geqslant 1} \sigma_\kappa^*(n)q^n$$

with $\zeta_p^*(\kappa) = L_p(1 - s, \chi)$ if $\kappa(a) = \langle a\rangle^s\chi(a)$, L_p the Kubota-Leopoldt p-adic L-function; and $\sigma_\kappa(n)^* = \sum_{d|n,(d,p)=1} \kappa(d)d^{-1}$.

Proposition 6. *If $\kappa(a) = \langle a\rangle^k\chi(a)$, then $G_\kappa(q) \in M_k(\Gamma_1(Np^l), \chi\omega^{-k})$, with l the level of the character χ, $k \in \mathbb{Z}$.*

Denote $E_\kappa = \frac{2G_\kappa^*(q)}{\zeta_p^*(\kappa)}$. We have $|E_\kappa^*(q) - 1|_{X_0} < 1$. Denote $E = E_{(1,0)} \in M_1(\Gamma_1(p), \omega^{-1})$ the Eisenstein series associated to the character $a \mapsto \langle a\rangle$. Denote moreover $e := \frac{E(q)}{E(q^p)}$, which is a weight 0 form, (i.e. a function).

Lemma 12. *$\forall \epsilon > 0$, there exists v such that e is defined on $X(v)$ and $|e - 1|_{X(v)} < \epsilon$.*

Proof. The proof relies on the inequality $|e - 1|_{X(0)} \leqslant p^{-1}$, and by the maximum modulus principle on $X(0)$ the norm "can't grow too big" if we just take a neighborhood. □

Definition 23. *Let $s \in B(1, p^{-1/p+1}) \subseteq \mathscr{W}$, we define e^s, and the operator u_s on $M_0(v)$ defined by*

$$u_s(h) = U_0(he^s)$$

As a result of Corollary 3, we have that, for $s = k \in \mathbb{Z}$, we have

$$\det(1 - Tu_k | M_0(v)) = \det(1 - TU_{p,k} | M_k(v))$$

And therefore, we can reduce our study of $U_{p,k}$ to study u_s. Moreover, we have the following:

Theorem 12. *Let $\mathscr{B} \subset \mathscr{W}$, then, there existas a unique $P(s,T) \in \mathbb{Q}_p[[s,T]]$ on $\mathscr{B} \times \mathbb{C}_p$, converging $\forall T \in \mathbb{C}_p$ and such that, for all $k \in \mathscr{B}$, we have*

$$P(k,T) = \det(1 - TU_{p,k} | M_k(v))$$

Now we define the spaces of modular forms:

Definition 24. *Let $\mathscr{B} \cong B(1, p^{-1/p+1}) \subseteq \mathscr{W}$ a connected affinoid of the weight space. We define $H^0(X(v) \times_K \mathscr{B}, \mathscr{O}_{X_v \times \mathscr{B}})$ the space of modular forms "à la Coleman".*

Observe that we can view s as a parameter and u_s as a family of operators. This gives an action

$$u_s : H^0(\mathscr{B} \times_K X(v), \mathscr{O}_{\mathscr{B} \times_K X(v)}) \cong M_0(v) \hat{\otimes} \mathscr{O}(\mathscr{B}) \to H^0(\mathscr{B} \times_K X(v), \mathscr{O}_{\mathscr{B} \times_K X(v)})$$

We should see the above definition as "spaces of overconvergent forms parametrized by a parameter in $\mathscr{B}$". The definition of Coleman relates to the usual spaces of overconvergent modular forms.

Theorem 13. *(See Buzzard, [2], Corollary 6.2) [1] Let $\kappa = x^k \epsilon$ be a classical weight, and $\psi : \mathscr{O}(\mathscr{B}) \to L$ the corresponding homomorphism, for L some finite extension of $\mathbb{Q}_p$. Then, multiplication by E_κ induces an isomorphism*

$$H^0(\mathscr{B} \times X(v), \mathscr{O}) \otimes_\psi L \cong H^0(X(v)_L, \underline{\omega}^{\otimes k})$$

That is, the fibres over a point in the weight space of Coleman's space of families of modular forms are isomorphic to the classical spaces of overconvergent modular forms.

Proof. (of Theorem 11, sketch) We have introduced all the ingredients needed for the proof of the main theorem. The rough idea is based on the fact that, if we restrict to a neighborhood of a weight k, the spectral theory of the $U_{p,s}$ operator doesn't change, so if for k there is a unique root $a_p(F)^{-1}$ of $P(k,T)$ of slope α, then $P(s,T)$ has also a unique root of slope α. □

Coleman's construction also allows to introduce modular forms of p-adic weights and p-adic families of modular forms. See B4 and B5 of Coleman's paper

Definition 25. *Let $F \in H^0(\mathscr{B} \times X(v), \mathscr{O})$ be a family of modular forms. Then FE^s, for $s \in \mathscr{B}$, is a weight s overconvergent modular form.*

It seems that Coleman and Buzzard follow some slightly diffenert terminology: Buzzard uses E_κ for the isomorphism between weight 0 and weight s, and Coleman uses E^s for $E = E_{(1,0)}$. It should be checked that these are the same

Coleman also shows ([5] B5.7.1) that forms live in p-adic families.

6.2 The eigencurve

We want to introduce a geometric object $\mathscr{D}$, called the eigencurve, which parametrizes systems of Hecke eigenvalues. This follows Sections 4 and 5 of [2]. First we introduce the spectral variety:

Definition 26. *We define the spectral variety as $\mathscr{Z} = V(P(s,T)) \subseteq \mathscr{B} \times \mathbb{A}^1$. It parametrizes inverse eigenvalues of $U_{p,s}$ by Corollary 3 and Theorem 12.*

We start by studying $\mathscr{Z}$. Let $B \subseteq \mathscr{B}$, $B = \mathrm{Max}(R)$ for R a Tate algebra. Consider $P(s,T)$ function on $B \times \mathbb{A}^1$, $\mathbb{A}^1 = \bigcup_{r \geqslant 0} \overline{B(0,p^r)}$. We define $Z_r \subseteq Z$ by the subvariety that is contained in $B \times \overline{B(0,p^r)}$. To define the eigenvariety we need a cover of the spectral variety. The following results follow closely Section 4 of [2].

[1] We remark that Buzzard is more careful about in which spaces do the weights live than we have been, because he includes the treatement of the Nebentypus.

Definition 27. *Let Z be the spectral variety and f, g the projection maps to B and $\mathbb{A}^1$, respectively. We denote by* Cov *the category whose objects are $Y \subseteq Z$ affinoid subdomains satisfying that, there exists some $X \subseteq B$ affinoid subdomain such that*

- $Y \subset Z_X := f^{-1}(X)$.
- $f : Y \to X$ *finite surjective.*
- $Z_X = Y \cup (Z_X \backslash Y)$ *is an admissible cover.*

Lemma 13. *$f_r : Z_r \to B$ is quasi-finite and flat.*

Proof. This is just Weierstrass preparation theorem. □

Corollary 4.

- *If $Y \subset Z$ is affinoid, the projection $f : Y \to B$ is quasi-finite and flat.*
- *If $Y \subset f^{-1}(X)$ and there is an integer d such that $f : Y \to X$ has all fibers with degree d, then f is finite and flat.*

Lemma 14. *For r fixed, $U_i = \{x \in B | \deg(f_r^{-1}(x)) \geqslant i\}$ is a union of affinoid subdomains, and U_i is empty for i large enough.*

Proof. Again Weierstrass preparation theorem. □

Lemma 15. *If $V \subseteq B$ be an affinoid such that $f_r^{-1}(V) \to V$ is finite of degree d, constant, then there exists some X strict neighborhood of V such that, maybe after extending the radius of convergence, we have $Y := f_s^{-1}(X) \supseteq f_r^{-1}(X)$ and $Y \in$ Cov.*

Theorem 14. Cov *is an admissible cover of Z.*

Proof. (Sketch) We know that Z is admissibly covered by the Z_r, so it suffices to prove that, for every Z_r, there is a finite collection of affinoids in Cov whose union contains Z_r. Let $U_i = \{x \in B | \deg(f_r^{-1}(x)) \geqslant i\}$ and $d = \max\{i \text{ such that } U_i \neq \varnothing\}$. Let $H(i)$ be the statement: "There is a finite set of affinoid subdomains of Z and a finite set $X_1, \dots X_{n(i)}$ of affinoid subdomains of B such that, for $1 \leqslant \alpha \leqslant n(i)$ we have $f : Y_\alpha \to X_\alpha$ is finite flat surjective, $Y_\alpha \in$ Cov, $f_r^{-1}(U_i) \subseteq \bigcup_{\alpha=1}^{n(i)} Y_\alpha$ and $\bigcup_{\alpha=1}^{n(i)} X_\alpha$ is a strict neighborhood of U_i in B. We have $H(1) \implies Z_r = \bigcup Y_\alpha$ and so that is what we want to prove. The proof follows by induction, c.f [2], Theorem 4.6. □

Now we construct the eigenvariety: The importance of the category Cov that we have defined is that its elements are the sets over which we can construct a Hecke algebra and therefore an eigenvariety, and the affinoids Y will glue to give the eigenvariety $\mathscr{D} \to \mathscr{Z}$.

Let M be a projective module over R, $U : M \to M$ a compact operator, and $Y \in$ Cov the admissible cover that we have constructed for $Z = V(P(s,T))$, for P the Fredholm determinant of U. Let X be image of this element Y of the cover. Then X is an affinoid subdomain of $\mathrm{Max}(R)$ and we denote $A = \mathscr{O}(X)$. Let's denote M_A for $M \hat{\otimes} A$, and P_A for the image of P in $A\{\{T\}\}$ (i.e. everywhere convergent power series), that is, the Fredholm determinant of P when we extend scalars to M_A, $P_A : M_A \to M_A$. We want to have a factorization $P(s,T) = Q(T)S(T)$, for Q a polynomial and S a unit.

Now by hypothesis $\mathscr{O}(Y)$ is a finite flat A-module, so it is projective of some rank d. Then let Q' be the characteristic polynomial of $T \in \mathscr{O}(Y)$, which is monic of degree d, and therefore gives

$$A[T]/Q'(T) \to \mathscr{O}(Y)$$

It can be seen that this map is surjective, and therefore an isomorphism, because both are locally free modules of the same rank d. Then we have $Q' = a_0 + \dots$, and $Q'(T) \mid F_A(T)$, and we can define $Q(T) = a_0^{-1} Q'(T)$. Observe that the quotient $P_A(T)/Q(T)$ is a unit, because Z_X is the disjoint union of Y and its complement. Then we are under the hypothesis of the hypothesis of Theorem 5, and so we have a decomposition $M_A = N \oplus F$, with N projective (free in this case) of rank d.

Now suppose that $\mathbb{T}$ is a commutative subalgebra of $\mathrm{End}(M_A)$, and if $\mathbb{T}$ commute with U, we get $\mathbb{T} \to \mathbb{T}_Y \subseteq \mathrm{End}(N)$. Then, we have

$$D(Y) = \mathrm{Spm}(\mathbb{T}_Y) \to Y$$

And we can glue $D(Y)$ together and get $\mathscr{D} \to \mathscr{Z} \to B^*$.

Observe that an L-point of $\mathscr{D}(Y)$ is a map $\mathbb{T} \to L$, sending $t \mapsto \lambda_t$, where λ_t is a root of the minimal polynomial of t. That is, an L-point is a system of eigenvalues.

In the particular case $M = H^0(X(v) \times B^\times, \mathscr{O})$, we get that $\mathscr{D}$ parametrizes systems of Hecke eigenvalues.

It can be shown that $\mathscr{D}$ is nice around classical points of non critical slope, that is, systems of Hecke eigenvalues for forms $f \in S_k(\Gamma_1(N) \cap \Gamma_0(p))$, primitive at N and of slope $< k - 1$.

Why the restriction to some B smaller subset of the weight space?.

Remark 23. *(Eigenvarieties) All this construction carries with the following data. Remark on links.*

7 Perfectoid modular curve

Our goal is to construct a modular curve $X = \varprojlim_n X_{\Gamma(p^n)}$ that parametrizes pairs $(E, \eta : T_pE \underset{\cong}{\longrightarrow} \mathbb{Z}_p^2)$. We should keep in mind the map $\mathbb{H} \to \mathbb{P}^1$ given by $\tau \mapsto [\tau : 1]$.

7.1 Perfectoid algebras

Definition 28. *A perfectoid field is a nonarchimedian complete field K of residue characteristic $p > 0$, non-discrete valuation group and such that the Frobenius endomorphism $\phi : K^0/(p) \to K^0/(p)$ is surjective.*

Definition 29. *For a perfectoid field K, we can define its tilt as $(K^0)^\flat = \varprojlim_\phi K^0/\omega$, with $\omega \in K^0$ such that $|p| \leqslant |\omega| < 1$. It can be shown that the definition does not depend on the choice of ω.*

Proposition 7. *1. If we eqquip $(K^0)^\flat$ with the inverse limit topology, then $(K^0)^\flat \cong \varprojlim_{x \mapsto x^p} K^0$ as topological multiplicative modules.*

2. There exists a multiplicative map $\# : (K^0)^\flat \to K^0$ given by projection to the first coordinate.

3. There exists $\omega^\flat \in (K^0)^\flat$ such that $(\omega^\flat)^\# = \omega$, and this allows to define $K^\flat := (K^0)^\flat[1/\omega^\flat]$.

4. $K^\flat$ is a perfectoid field with valuation $|.|^\flat$ given by $|x|^\flat = |x^\#|$.

5. We have an isomorphism $K^0/(\omega) \cong (K^0)^\flat/(\omega^\flat)$.

Example 7. *Probably the best known example of this setting are the rings $\mathscr{O}_{\mathbb{C}_p}$ and $\mathscr{O}_{\mathbb{C}_p}^\flat$ used in p-adic Hodge Theory.*

Let's fix K a perfectoid field, $\omega, \omega^\flat$ as above. Let $\mathfrak{m} \subseteq K^0$ be the maximal ideal.

Definition 30. *We say that a K^0-module M is almost zero if $\mathfrak{m}M = 0$. We can define the category of almost zero K^0-modules, denotes $(K^0)^a$-Mod, to be the localized category $K^0 - \mathrm{Mod}/$almost zero $K^0 - \mathrm{Mod}$. We denote M^a the image of M in the localized category $(K^0)^a$-Mod.*

Observation 1. *$(K^0)^a$-Mod is a tensor category via $M^a \otimes N^a$. Since we have a tensor product, we can define the category of $(K^0)^a - \mathrm{alg}$, and for $R \in (K^0)^a - \mathrm{alg}$, we can define R-algebras.*

Definition 31. *1. A perfectoid K-algebra is a Banach K-algebra R such that*

- *R^0 is open and bounded.*
- *$\phi : R^0/\omega \to R^0/\omega$ is surjective.*

2. A perfectoid $(K^0)^a$-algebra is a ω-adicallt complete flat $(K^0)^a$-algebra A such that $\phi : A/\omega^{1/p} \to A/\omega$ is an isomorphism.

3. A perfectoid $(K^0/\omega)^a$-algebra is a flat $(K^0/\omega)^a$-algebra $\overline{A}$ such that $\phi : \overline{A}/\omega^{1/p} \to \overline{A}$ is an isomorphism.

Theorem 15. *We have a chain of equivalences of categories*

$$K - \mathrm{perf} \to (K^0)^a - \mathrm{perf} \to (K^0/\omega)^a - \mathrm{perf} \to \left((K^0)^\flat/\omega^\flat\right)^a - \mathrm{perf} \to \left((K^\flat)^0/\omega^\flat\right)^a - \mathrm{perf} \to K^\flat - \mathrm{perf}$$

7.2 Construction of the anticanonical tower

Let $N \geqslant 3$ and let p be a prime such that $p \nmid N$. Let $K^p \subset GL_2(\mathbb{A}_f^+)$ compact open open such that $K^p \subseteq \{g \in GL_2(\hat{\mathbb{Z}}^p)$ such that $g \equiv 1 \mod N\}$. Let $K_p \subset GL_2(\mathbb{Z}_p)$ be either $GL_2(\mathbb{Z}_p)$, $\Gamma_0(p^n)$, $\Gamma_1(p^n)$ or $\Gamma(p^n)$. Then the moduli problen of elliptic curves with level K^pK_p-structure is representable.

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